

Telink

Amazon Sidewalk

User Manual

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Brief

This document provides a user manual for the Amazon Sidewalk.

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Revision History

Version	Revisions
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V1.0.0	Initial Release
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1 Overview

Amazon Sidewalk is a secure community network protocol for the Internet of Things (IoT) launched by Amazon. For more details, please visit Telink's official website: [Amazon Sidewalk](#).

The protocol supports three wireless transmission methods:

- Amazon Sidewalk SubG-FSK, referred to as SubG-FSK
- Amazon Sidewalk SubG-CSS (based on LoRa[®] technology), referred to as SubG-CSS
- Bluetooth[®] Low Energy, referred to as BLE

This manual helps users build a Sidewalk development environment from scratch using the Telink Sidewalk Development Kit (AIOT-DK1+ML3238S). It includes hardware introduction, SDK import & compilation, firmware flashing, AWS configuration and demo verification. Below is the core data transmission process:



Figure 1.1: Data Transmission Process

Related Documents & Tools

- [Amazon Sidewalk Quick Start Guide](#)
- [AIOT-DK1 User Manual](#)
- [Telink Development Tools Download](#)
- Documents Related to Telink Development Tools
 - [Telink VS Code Extension User Guide](#)
 - [Burning and Debugging Tool \(Windows\) User Guide](#)
 - [Burning and Debugging Tool \(Linux & MAC\) User Guide](#)
 - [Telink IoT Studio User Guide](#)

2 Hardware Overview

2.1 Hardware Preparation

- Sidewalk development kit
 - Telink AIOT-DK1 development motherboard
 - Telink ML3238S development daughterboard (for Amazon Sidewalk)
- USB Type-C cable (for power supply and firmware downloading)
- Sidewalk Gateway (Echo 4 used in this demo). Refer to the [List of Sidewalk Gateways](#) for more options.
- Wi-Fi router (Spitz AX (GL-X3000, Wi-Fi 6, VPN-supported, dual SIM))
 - The gateway needs Wi-Fi network access to connect to the AWS Cloud.
- (Optional) UART-USB adapter + jumper wires (for log viewing)

2.2 Sidewalk Development Kit

The Amazon Sidewalk development kit is a comprehensive platform designed for Amazon Sidewalk applications. It consists of Telink's AIOT-DK1 general-purpose development motherboard and dedicated ML3238S daughterboard. Supporting BLE, SubG-FSK, and SubG-CSS wireless communication protocols, the kit integrates functions including programming and debugging, functional development and demonstration, RF testing, and power consumption measurement, enabling rapid device prototyping and performance validation.



Figure 2.1: Amazon Sidewalk Development Kit

2.2.1 AIOT-DK1 Development Motherboard

The AIOT-DK1 is a multifunctional general development motherboard designed by Telink. It is used in conjunction with various Telink modules to perform functional development, verification, and demonstrations, serving as the core hardware platform for the entire kit. For the complete AIOT-DK1 hardware manual, pin

definitions, circuit schematics, and usage specifications, please refer to the official documentation: [AIOT-DK1 User Manual](#).

Core Functions and Features:

- 3 types of onboard sensors, multiple LEDs, multiple buttons, and hardware reset;
- 3 onboard mikroBUS™ sockets for flexible functionality expansion;
- Onboard burning and debugging module supporting direct firmware burning and online debugging;
- Equipped with 3 USB Type-C ports for power supply, burning, and debugging;
- Supports module power consumption testing.

The key features of **ML3238S + AIOT-DK1** are described in the following sections and listed in the table below.

No.	Features	ML3238S-G	Instruction
1	Digital microphone	-	Toggle SW7 left
2	SDM JACK	-	Toggle SW8 down
3	Camera	-	Toggle SW17 left
4	Module reset - SW2	Y	-
5	Normal button - SW4	Y	-
6	Normal button - SW5	Y	-
7	Normal button - SW6	Y	-
8	LED - D1 (White)	Y	-
9	LED - D2 (Green)	Y	-
10	IR	-	-
11	USB Full-speed	-	-
12	USB High-speed	-	-
13	Pressure sensor	Y	-
14	Relative humidity and temperature sensor	Y	-
15	3D accelerometer, 3D gyroscope, 3D magnetometer sensor	Y	-

2.2.2 ML3238S Development Daughterboard

The ML3238S is a dedicated development daughterboard designed specifically for Amazon Sidewalk. It should be used in conjunction with the AIOT-DK1 development motherboard and cannot operate independently. The ML3238S serves as the key platform for implementing Amazon Sidewalk-related functions.

The core hardware configuration is as follows:

- Onboard KGM100XB wireless communication module, providing wireless communication support for Sidewalk functionality;
- Integrated 2.4 GHz & 900 MHz dual-band chip antenna, ensuring stable wireless signal transmission;
- Equipped with an RF coaxial connector with a switch, supporting external antenna expansion and RF performance testing.

The following provides the complete schematic of the ML3238S, hardware connections to the AIOT-DK1 motherboard, pin assignments, and debugging considerations.

2.2.2.1 ML3238S Schematic Design

The ML3238S schematic centers on U1 (module interface) and U2 (KGM100XB) as shown in the figure below, connecting directly to the J2 connector on the AIOT-DK1 motherboard. It supports pin multiplexing, RF switching, and a dual-antenna design.

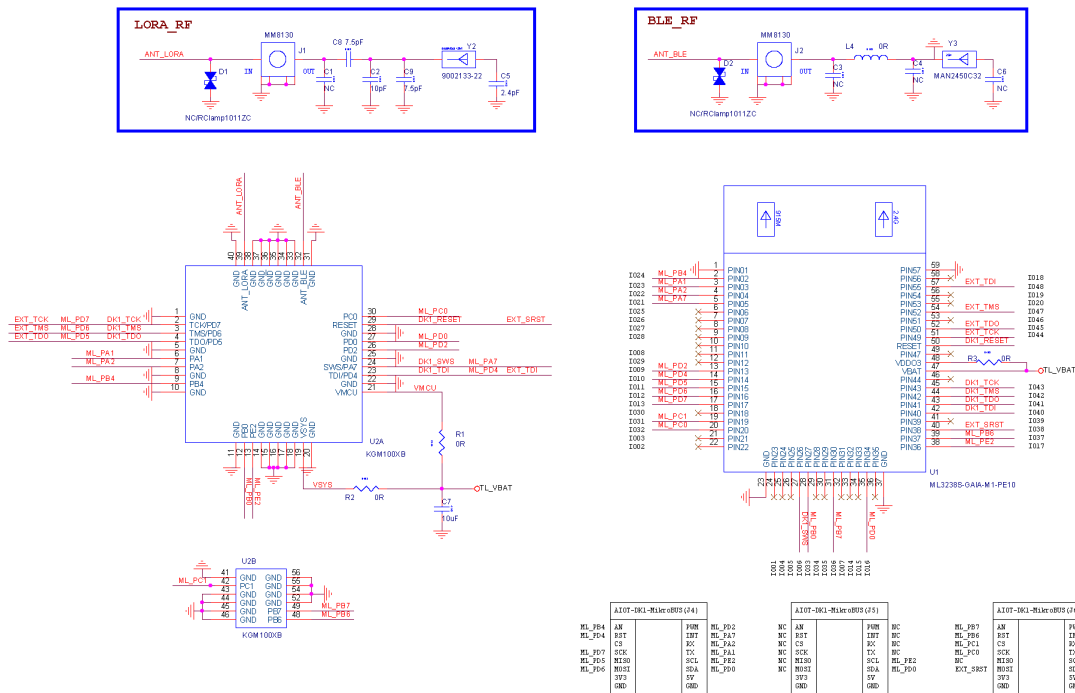




Figure 2.3: J4/J5/J6 Connection with ML3238S

2.2.2.2 U1 Pin Assignment and Multiplexing Description

U1 serves as the external interface of the ML3238S module and corresponds to AIOT-DK1's J2 connector. To enhance flexibility, some GPIOs support function multiplexing and can be configured by the user according to application scenarios.

JTAG & All-Function GPIO Multiplexing:

PD4/PD5/PD6/PD7 support both All-Function GPIO and JTAG debugging modes:

- ML_PD4/PD5/PD6/PD7: Connect to mikroBUS (J4) and can be configured as standard GPIOs for peripheral expansion.

- DK1_TCK/TMS/TDO/TDI: Connect to the AIOT-DK1 onboard programmer for onboard JTAG debugging.
- EXT_TCK/TMS/TDO/TDI: Connect to mikroBUS (J6) for external debugging using standard debuggers such as J-Link.

Note:

- For All-Function GPIO definitions, please refer to the TL3238E1 chip datasheet (contact Telink sales or FAE for this document).

SWS & GPIO Multiplexing:

The chip supports a single-wire communication interface (Single Wire Slave). The SWM/SWS protocol is a Telink single-wire master/slave protocol with a maximum data rate of 2 Mbps, enabling burning and debugging via the onboard programmer:

- DK1_SWS: Connects to the AIOT-DK1 onboard programmer for firmware burning and online debugging.
- ML_PA7: The SWS pin is multiplexed as a standard GPIO and connected to the mikroBUS (J4), allowing it to be used as a general-purpose I/O.

Reset & SRST Multiplexing:

- DK1_RESET: Connects to the AIOT-DK1 reset button (SW2) and supports hardware reset.
- EXT_SRST: Connects to the JTAG standard interface SRST, connecting with the J-Link reset pin for tool-controlled reset.

2.2.2.3 RF and Antenna Design

RF Switch Connector (MM8130): The Murata MM8130 RF coaxial connector (SWF series, SMD package) is used:

- Features: 50Ω impedance, supports 6 GHz bandwidth, built-in test switch.
- Function: Allows disconnection of the onboard antenna and connection to external instruments for RF debugging, calibration, and testing.
- Note: Corresponding Murata test probe (for conducted RF testing) is MXHS83QE3000 or MXHT83QE3000 (included in Sidewalk Development Kit).

Chip Antenna

- 2.4 GHz Chip Antenna: Dedicated to BLE communication.
- 900 MHz Chip Antenna: Dedicated to SubG-FSK/SubG-CSS long-range communication.
- Compact SMD Design: Facilitates product integration and development verification.

2.2.3 KGM100XB Wireless Communication Module

The KGM100XB is a wireless module designed for Amazon Sidewalk. It supports dual-link communication featuring BLE, SubG-FSK, and SubG-CSS, making it ideal for low-power, long-range devices in IoT and smart home applications.

Core chip components:

- TL3238E1 SoC: The main controller, responsible for BLE communication and MCU functions.
- SX126x Transceiver: A high-performance LoRa/FSK transceiver responsible for long-range, low-power communication.

2.2.4 Onboard Programmer

The onboard programmer is a dedicated burning and debugging unit integrated onto the AIOT-DK1 motherboard, enabling firmware burning and hardware debugging without the need for an external debugger. Key advantages include:

- Integrated onboard design eliminates the need to purchase an external debugger, simplifying the development process.
- Supports three debugging modes: Single-Wire (Swire), SDP, and JTAG.
- Equipped with status indicators for intuitive display of burning and debugging status.
- Supports one-click program burning, Flash erasure, chip activation, and memory read/write operations.

This kit should be used in conjunction with the Burning and Debugging Tool (BDT).

- Telink official burning and debugging software for Windows, Linux, and macOS.
- Supports firmware download, Flash erasure, security encryption and decryption, variable debugging, and log viewing.
- Official tool download link: [Telink | Development Tools](#)
- Detailed user guide (including GUI and command line): [Burning and Debugging Tool \(Windows\)](#)

3 AWS Cloud Environment Setup

Go through the following steps to deploy the Amazon reference app and provision your device.

Your outcome shall be a working **web application**.

3.1 Clone the AWS Sample App

Clone the web application's repository:

```
git clone https://github.com/aws-samples/aws-iot-core-for-amazon-sidewalk-sample-app.git
cd aws-iot-core-for-amazon-sidewalk-sample-app
```

3.2 Set Up Python Environment

Setting a virtual Python environment is the preferred way of proceeding in this document.

Install pyenv, referring to [Simple Python Version Management: pyenv](#), according to your system (Linux or Windows).

Download preferred Python version:

```
pyenv install 3.9.18
```

Set up and activate the virtual environment for the AWS repository (still in the AWS repository folder).

```
pyenv virtualenv 3.9.18 sidewalk-aws-39
pyenv local sidewalk-aws-39
```

Check whether it is working by verifying the Python version.

```
python -V
```

It should show 3.9.18.

Use pip (under a virtual environment) to install all needed packages.

```
python -m venv sample-app-env
source sample-app-env/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python -m pip install pyjwt -t ./ApplicationServerDeployment/lambda/authLibs
```

3.3 Set up AWS Account

Set up an AWS account (the account itself is free to create, and the demo does not require any paid features).

Note:

- The word "Console" used by Amazon refers to the web interface controlling AWS accounts. This is not a console in "normal" embedded understanding.

After logging in to the AWS root console, create an IAM user.

Open the IAM service (use the search bar at the top of the page if IAM is not visible), then open the "IAM users" under "Access Management". Click the "Create user" button on the right.

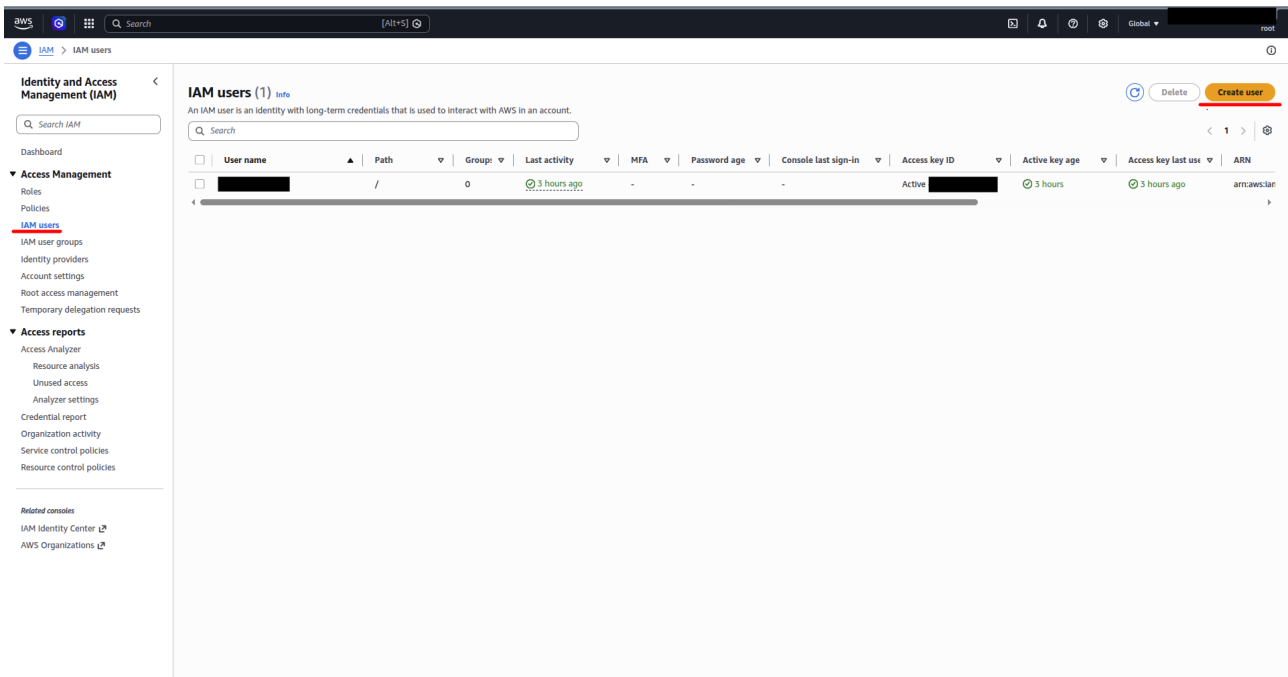


Figure 3.1: Create a New User

Enter a user name (for example, Demo_User) and click "Next".

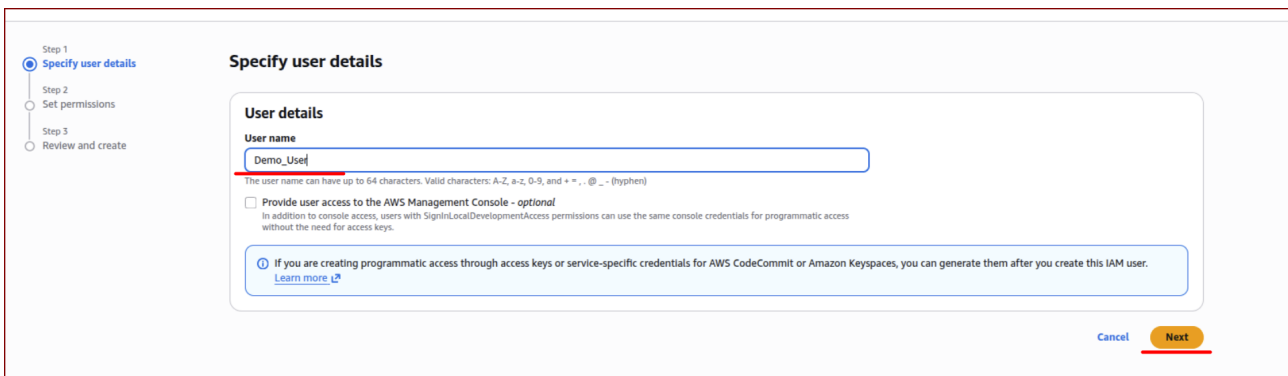


Figure 3.2: Specify User Details

Set permissions for your newly created user.

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1492)

Choose one or more policies to attach to your new user.

Filter by Type: All types 53 matches

Policy name	Type	Attached entities
☒ AdministratorAccess	AWS managed - job function	1
☐ AdministratorAccess-Amplify	AWS managed	0
☐ AdministratorAccess-AWSElasticBeanstalk	AWS managed	0
☐ AIOpsConsoleAdminPolicy	AWS managed	0
☐ AmazonAPIGatewayAdministrator	AWS managed	0
☐ AmazonNimbleStudio-StudioAdmin	AWS managed	0
☐ AmazonSageMakerAdmin-ServiceCatalogProdu...	AWS managed	0
☐ AmazonSageMakerHyperPodObservabilityAdm...	AWS managed	0
☐ AmazonSecurityLakeAdministrator	AWS managed	0
☐ AmazonWorkSpacesAdmin	AWS managed	0

Figure 3.3: Set Permissions

Click "Next", then click "Create user" at the bottom of the next page.

Note:

- To speed up the setup process, assign Administrator permissions. For additional details on AWS permission configuration, please refer to [Appendix 1](#).

Generate an access key for the IAM user:

From the root console, navigate to the IAM user's panel:

root console -> IAM (already invoked or use "search" bar) -> Access Management -> Users -> click on user's name

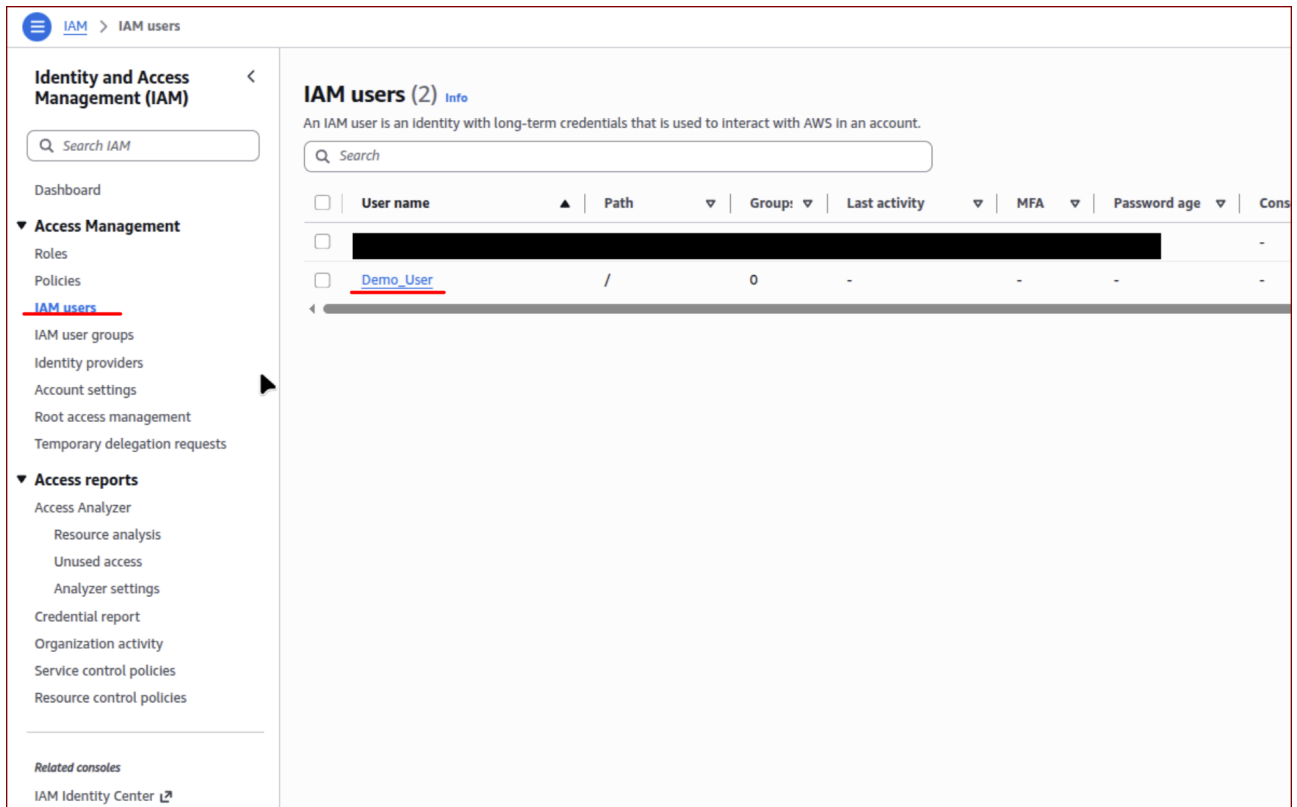


Figure 3.4: Access Management Page

Click "Create access key" and store the resulting credentials pair:

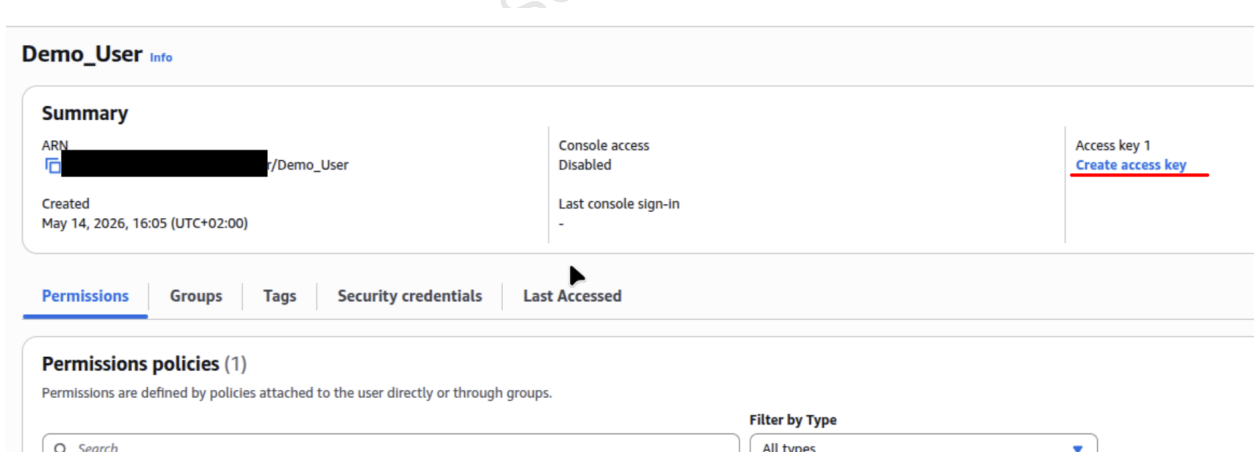
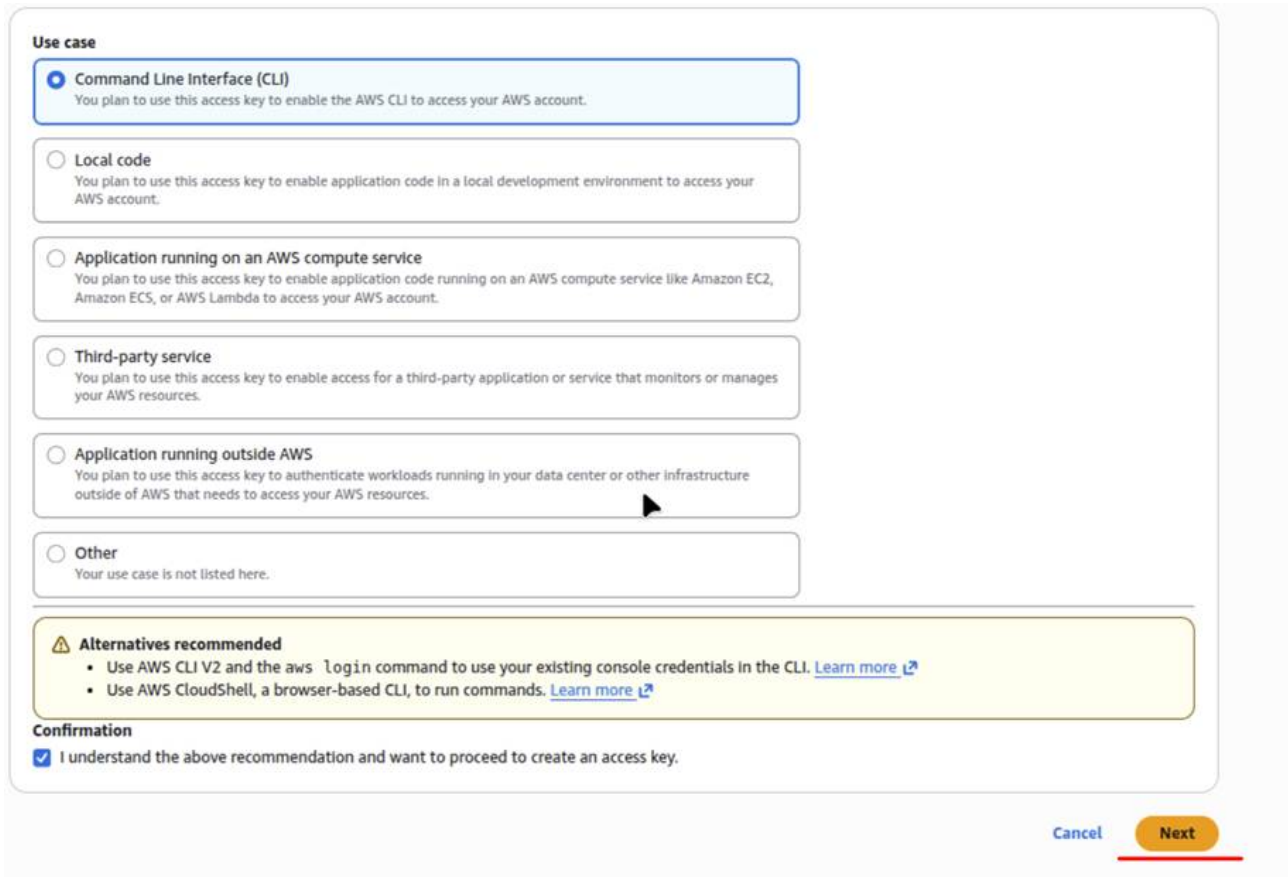


Figure 3.5: Create Access Key

Note:

- Since the use of access keys has not been recommended by Amazon for some time, you must ignore the upcoming warnings for the purposes of this demo:



Use case

☒ **Command Line Interface (CLI)**
You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ **Local code**
You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ **Application running on an AWS compute service**
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

☐ **Third-party service**
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.

☐ **Application running outside AWS**
You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.

☐ **Other**
Your use case is not listed here.

Alternatives recommended

- Use AWS CLI V2 and the `aws login` command to use your existing console credentials in the CLI. [Learn more](#)
- Use AWS CloudShell, a browser-based CLI, to run commands. [Learn more](#)

Confirmation

☒ I understand the above recommendation and want to proceed to create an access key.

[Cancel](#) [Next](#)

Figure 3.6: Use Case

Key ID should look like

AKIAIOSFODNN7EXAMPLE

and the secret key should look like:

wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY

Store these in a safe place for further use.

3.4 Configure AWS CLI

`aws` is a command-line tool to interact with Amazon servers. First, you need to configure it. Go back to the terminal with `aws-iot-core-for-amazon-sidewalk-sample-app` used in the sections [Clone the AWS Sample App](#) and [Set Up Python Environment](#).

aws configure

Follow the prompts and provide credentials for your IAM user. Use the key obtained in the steps above.

Please note that Amazon Sidewalk is currently available only in the United States. Select an appropriate region ("us-east-1" has been tested successfully).

```
$ aws configure
AWS Access Key ID []: KAI[EXAMPLE]S3T
AWS Secret Access Key []: Dj7[EXAMPLE][EXAMPLE]qfznt
Default region name []: us-east-1
Default output format [None]:
```

3.5 Deploy the Sensor Monitoring Web App

Edit the config.yaml file:

```
*HARDWARE_PLATFORM* set to anything you like, NORDIC seems to be OK
*USERNAME* change to preferred user name for demo web app
*PASSWORD* set password for demo web app
```

Run deploy script:

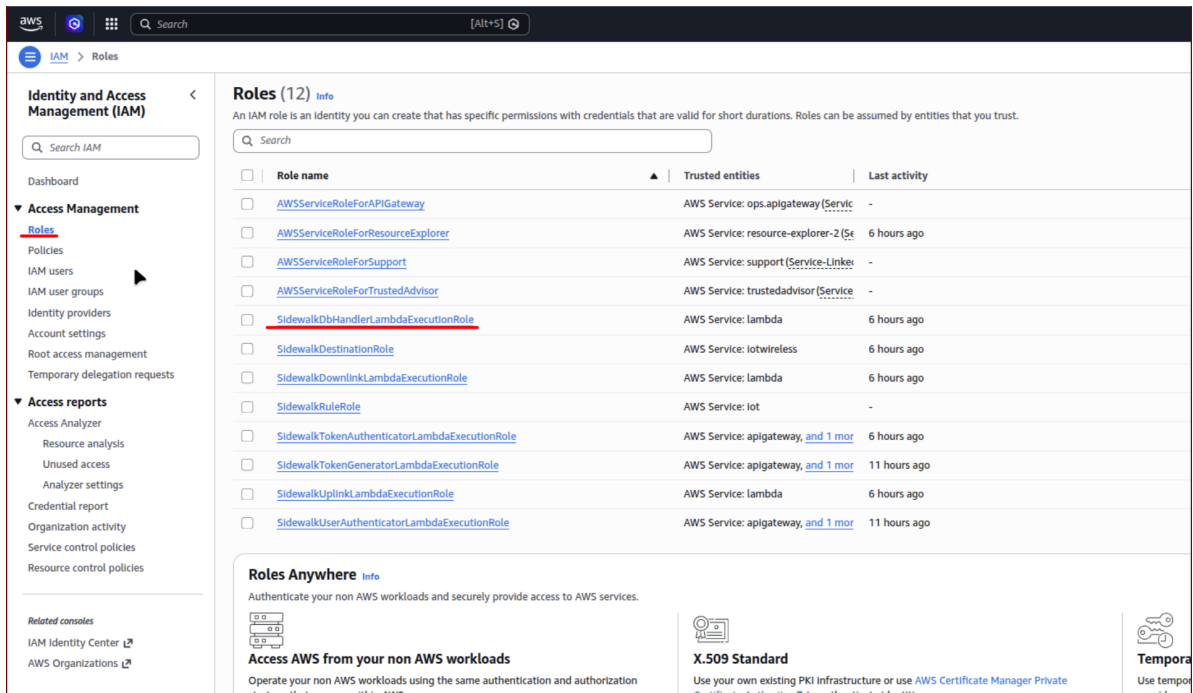
```
python ApplicationServerDeployment/deploy_stack.py
```

It may take several minutes for the script to finish.

3.6 Add Privileges to the Database Handler

Although your IAM user has Administrator permissions, AWS also uses so-called "Roles", which require their own permissions.

Navigate to the IAM service. This time, instead of selecting "IAM users", select "Roles", then choose "Side-walkDbHandlerLambdaExecutionRole".



Roles (12) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Role name	Trusted entities	Last activity
AWSServiceRoleForAPIGateway	AWS Service: ops.apigateway (Service-Linker)	-
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Service-Linker)	6 hours ago
AWSServiceRoleForSupport	AWS Service: support (Service-Linker)	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linker)	-
SidewalkDbHandlerLambdaExecutionRole	AWS Service: lambda	6 hours ago
SidewalkDestinationRole	AWS Service: iotwireless	6 hours ago
SidewalkDownlinkLambdaExecutionRole	AWS Service: lambda	6 hours ago
SidewalkRuleRole	AWS Service: iot	-
SidewalkTokenAuthenticatorLambdaExecutionRole	AWS Service: apigateway, and 1 more	6 hours ago
SidewalkTokenGeneratorLambdaExecutionRole	AWS Service: apigateway, and 1 more	11 hours ago
SidewalkUplinkLambdaExecutionRole	AWS Service: lambda	6 hours ago
SidewalkUserAuthenticatorLambdaExecutionRole	AWS Service: apigateway, and 1 more	11 hours ago

Roles Anywhere Info

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization as your AWS workloads.

X.509 Standard

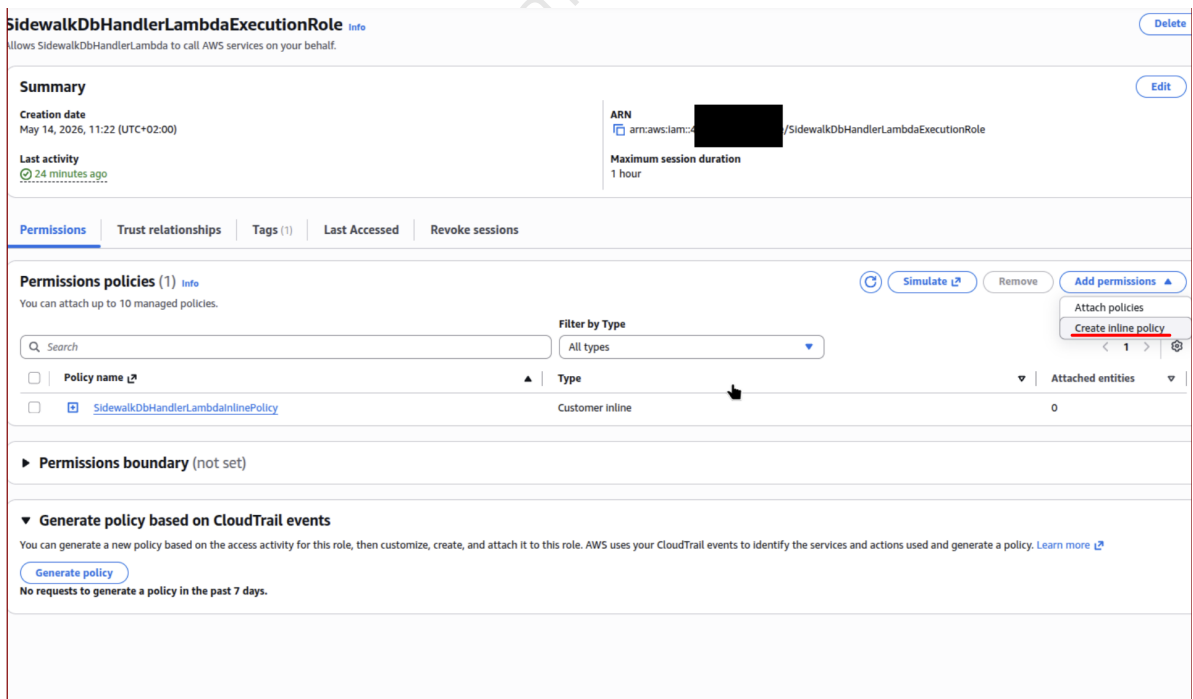
Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

Temporary Credentials

Use temporary credentials to access AWS services.

Figure 3.7: Roles

On the Role details page, click "Create inline policy" from the "Add permissions" menu.



SidewalkDbHandlerLambdaExecutionRole Info

Allows SidewalkDbHandlerLambda to call AWS services on your behalf.

Summary

Creation date: May 14, 2026, 11:22 (UTC+02:00)

Last activity: 24 minutes ago

ARN: [arn:aws:iam::\[redacted\]:role/SidewalkDbHandlerLambdaExecutionRole](#)

Maximum session duration: 1 hour

Permissions | Trust relationships | Tags (1) | Last Accessed | Revoke sessions

Permissions policies (1) Info

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
SidewalkDbHandlerLambdaInlinePolicy	Customer inline	0

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

Figure 3.8: Create Inline Policy

Select "JSON" and paste the following:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowScanOnSpecificIndex",
      "Effect": "Allow",
      "Action": "dynamodb:Scan",
      "Resource": [
        "arn:aws:dynamodb:us-east-1:[account_ID]:table/SidewalkMeasurements",
        "arn:aws:dynamodb:us-east-1:[account_ID]:table/SidewalkMeasurements/index/*"
      ]
    }
  ]
}
```

Replace `[account_ID]` with your account ID (a long numeric value that can be found in your account profile).

Then click "Next".

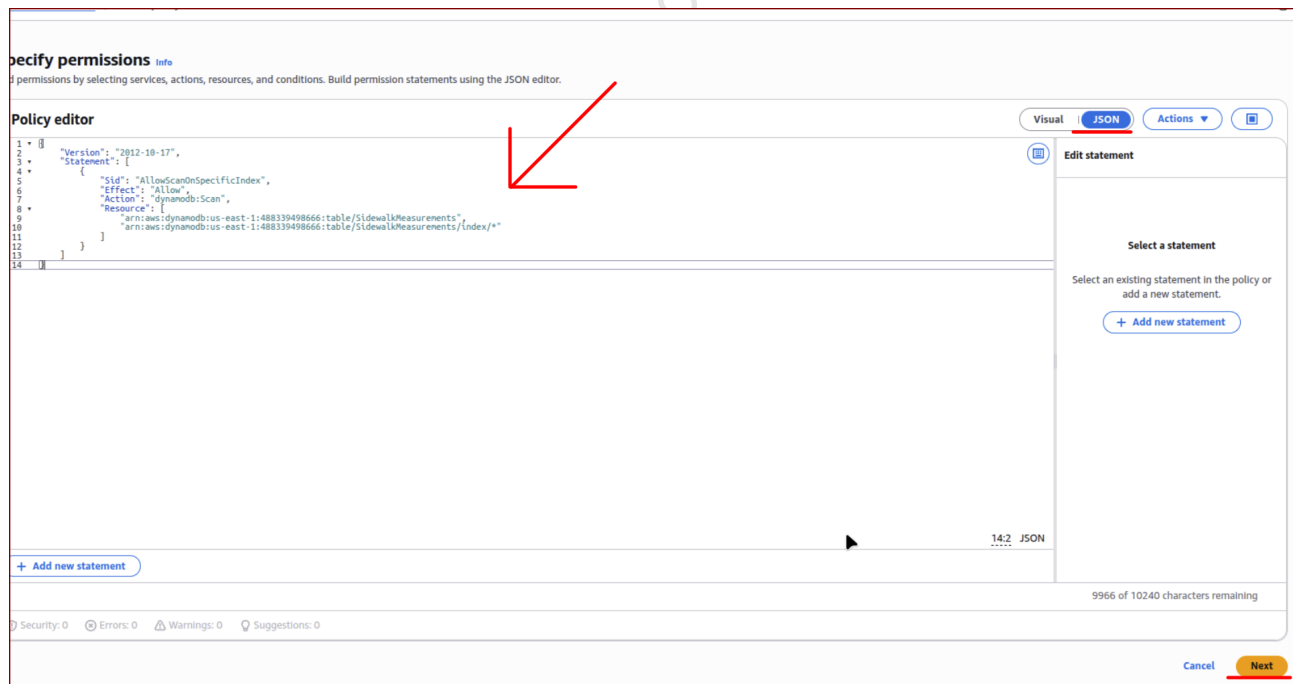


Figure 3.9: Policy Editor

Enter a meaningful name for the policy (for example, demoDynaDB), then click "Create policy".

Review and create
[Info](#)

Review the permissions, specify details, and tags.

Policy details

Policy name
 Enter a meaningful name to identify this policy.

Maximum 128 characters. Use alphanumeric and "+", "@", "-" characters.

Permissions defined in this policy
[Info](#)
[Edit](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM Identity (user, user group, or role), attach a policy to it.

Allow (1 of 469 services)
[Show remaining 468 services](#)

Service	Access level	Resource	Request condition
DynamoDB	Limited: Read	Multiple	None

[Cancel](#)
[Previous](#)
[Create policy](#)

Figure 3.10: Policy Name

Go back to the Role details page and select “Attach policies” from the “Add permissions” menu.

ionRole

SidewalkDbHandlerLambdaExecutionRole
[Info](#)
[Delete](#)

Allows SidewalkDbHandlerLambda to call AWS services on your behalf.

Summary
[Edit](#)

Creation date
 May 14, 2026, 11:22 (UTC+02:00)

ARN
 arn:aws:iam::[redacted]:role/SidewalkDbHandlerLambdaExecutionRole

Last activity
 8 minutes ago

Maximum session duration
 1 hour

[Permissions](#)
[Trust relationships](#)
[Tags \(1\)](#)
[Last Accessed](#)
[Revoke sessions](#)

Permissions policies (1)
[Info](#)
[Simulate](#)
[Remove](#)
[Add permissions](#)

You can attach up to 10 managed policies.

Filter by Type
 All types

☐	Policy name	Type	Attached entities
☐	SidewalkDbHandlerLambdaInlinePolicy	Customer inline	0

Permissions boundary (not set)

Generate policy based on CloudTrail events
 You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)
[Generate policy](#)
 No requests to generate a policy in the past 7 days.

Figure 3.11: Attach Policies

Select “AmazonDynamoDBFullAccess” and click “Add permissions”.

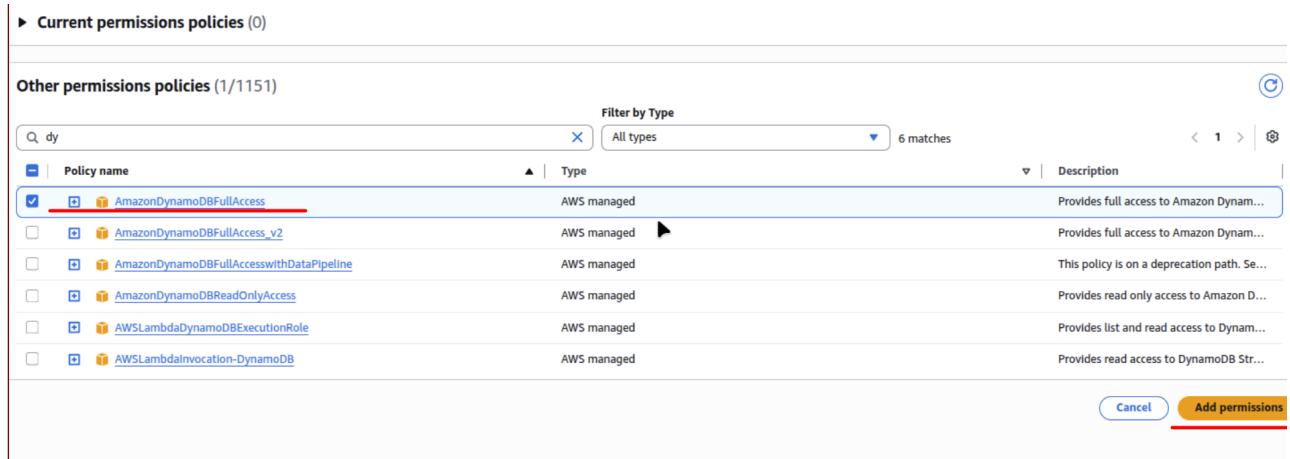


Figure 3.12: Other Permissions Policies

Go back to the web application and refresh the page. You should see either a temperature measurement plot or a blank panel displaying “No Data”, but you should no longer see the “Error loading measurements, try again” message.

3.7 Ensure the Web App Is Up and Running

Open

```
aws-iot-core-for-amazon-sidewalk-sample-app/config.yaml
```

and look for:

```
USERNAME: awesome_user
PASSWORD: very_top_secret_password

WEB_APP_URL: blah_blah.cloudfront.net
```

Open a web browser of your choice and use `WEB_APP_URL` and the credentials. See the web app, but no devices yet.

4 Sidewalk SDK Overview

This section details all necessary settings to use Telink's Sidewalk EVK (AIOT-DK1 + ML3238S) in running two different applications:

- Sensor Monitor and SID DUT (Sections [Prerequisites](#) through [Flash the Board](#)).
- Later sections, [Test the Sidewalk Demo](#) and [Test the DUT Demo](#), provide the necessary test steps for the two different applications. Please note the following:

Please note the following:

- The AWS environment configuration steps, which are necessary to support both the Sidewalk Sensor and SID DUT tests, are carefully explained.
- Note that all necessary steps to compile the SID DUT are provided. However, the Sidewalk Sensor binary is currently available only as a download from here: [sidewalk_sensor_monitoring.bin](#)

4.1 Prerequisites

4.1.1 Hardware

There are four important components involved:

- AWS cloud infrastructure running a demo web app
 - Refer to the tutorial [Build a Basic AWS Web Application](#), which uses AWS Amplify to build a full-stack web application.
- Sidewalk gateway
 - Echo 4 (used in this demo) or other compatible Amazon Sidewalk gateways.
 - For detailed gateway models and their supported communication protocols (BLE, SubG-CSS, SubG-FSK), refer to the [List of Sidewalk Gateways](#)
- Sidewalk end device
 - Demonstration with Telink Sidewalk Kit (AIOT-DK1 + ML3238S)
 - * AIOT-DK1 motherboard: Refer to the [AIOT-DK1 User Manual](#), which supports multiple modules, integrates 3 sensors, 3 mikroBUS interfaces, and supports onboard burning.
 - UART-USB converter for logs (optional)
 - Telink Programmer
 - Arduino jumper cables (male-to-female jumper cables are used for the USB-to-UART converter, and female-to-female jumper cables are used for the Telink Programmer)
- OpenVPN-capable Wi-Fi router (required outside the US, Canada, and Mexico)

As of March 2026, direct access is limited to the United States, Canada and Mexico. Users in other regions must connect to the server via a VPN.

4.1.2 Software

Install the required toolchains, programming, and debugging tools. Official Tools Download Link: [Telink Development tools](#)

- Telink VS Code Extension
 - Telink VS Code Extension is an all-in-one embedded development tool specifically built for Telink chips. Deeply integrated into the ecosystem of Visual Studio Code (VS Code) and its derivative AI editors, it is designed to provide developers with an efficient full-lifecycle development experience covering environment configuration, project creation, coding, compilation, debugging, and firmware burning.
 - Refer to the [Telink VS Code Extension User Guide](#).
- Burning and Debugging Tool (BDT)
 - The Burning and Debugging Tool is applicable to the application development of the Telink series SoC. During the SDK development process, the firmware can be directly downloaded to the target development board via USB mode or Telink Programmer. The main functions include: erasing Flash sectors, firmware download, activating the MCU, accessing memory spaces including Flash/Core/Analog/OTP, reading/writing global variables, and viewing USB logs.
 - Refer to the [Burning and Debugging Tool User Guide](#).
- Python 3.9+
- [AWS CLI](#)
- Git

Info:

- For the installation and usage of BDT, VS Code, and Git, please refer to their respective user guides. The configurations of Python and the AWS CLI are described in the following sections.

4.2 Set Up the Sidewalk Gateway Outside the US

Until March 2026, Amazon Sidewalk is available only in North America (the US, Canada, and Mexico). Users in the US, Canada, and Mexico require only an Amazon Sidewalk gateway, whereas users in other regions should refer to this section.

To develop Sidewalk solutions outside North America, the gateway device must be “tricked” into being USA-located. This can be done by VPN capable Wi-Fi router. “GL.Inet GL-MT300N-V2 Router Mango Smart Mini” device with a Nord VPN account is proven to work.

Amazon suggests using open VPN protocol for this network purpose. Open VPN provided by Nord VPN has been used in this demo with success. No other VPN provider nor VPN protocol has been tested.

Amazon Sidewalk uses 902-928 MHz for sub-GHz operation. In the EU, this frequency range collides with the LTE band, which is strictly protected by UKE (Office for Electronic Communication) in Poland and its equivalent government agencies in other European countries. To develop sub-GHz capabilities of the Sidewalk (which are out of scope for this demo), a coaxial cable is recommended instead of using over-the-air transmission.

Note:

- European availability is planned for 2026, but has not yet been announced. Messaging for EU (European Union) audiences should avoid implying current coverage. Coverage maps are available for this region: [Sidewalk Coverage](#).

A shielded room/cage should be used for outside-US operation of sub-GHz modes (FSK, LoRA). To reduce development cost, an RF shielded cage can be avoided. The Echo 4 device can be converted for cable operation with u.FL -> SMA "pigtail". Using a 40~60dB attenuator is recommended to avoid damage to the radios. This approach, however, requires some basic RF and hardware handling skills and is not officially recommended.

Refer to the Amazon documentation to set up the Echo 4 or other compatible sidewalk gateway. It is a good starting point to follow this link: [Setup an Amazon Echo as a Sidewalk Gateway in a Non-US Location](#).

4.3 Build the Firmware

GitHub Repository: https://github.com/telink-semi/tl_amazon_sidewalk

Execute the following commands to clone the repository and switch to the specified branch:

```
$ git clone https://github.com/telink-semi/tl_amazon_sidewalk
$ cd tl_amazon_sidewalk
```

Start VS Code:

```
$ vscode .
```

Open Telink extension in VS Code and find "Project Outline" sidebar. Use the "convert" button if projects are not visible.

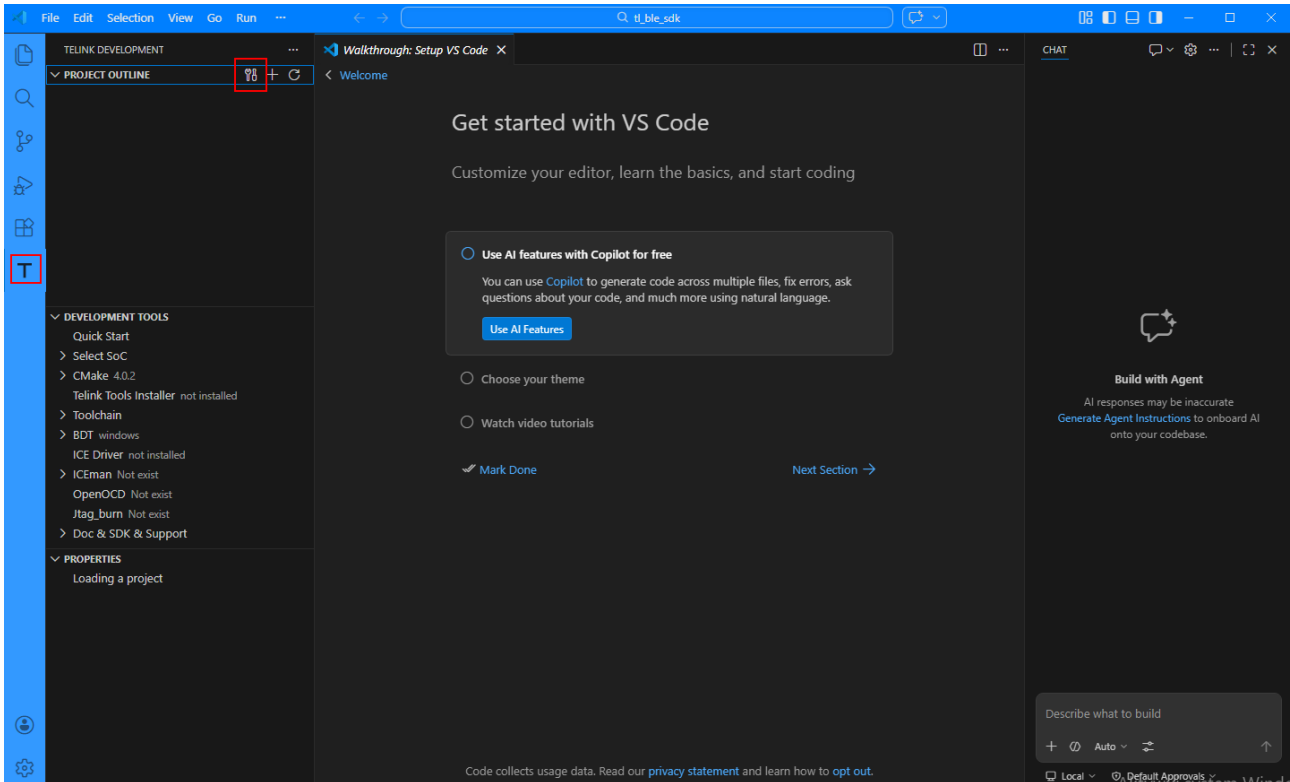


Figure 4.1: outline_and_convert

Select proper SoC (TL3238E1 in this case)

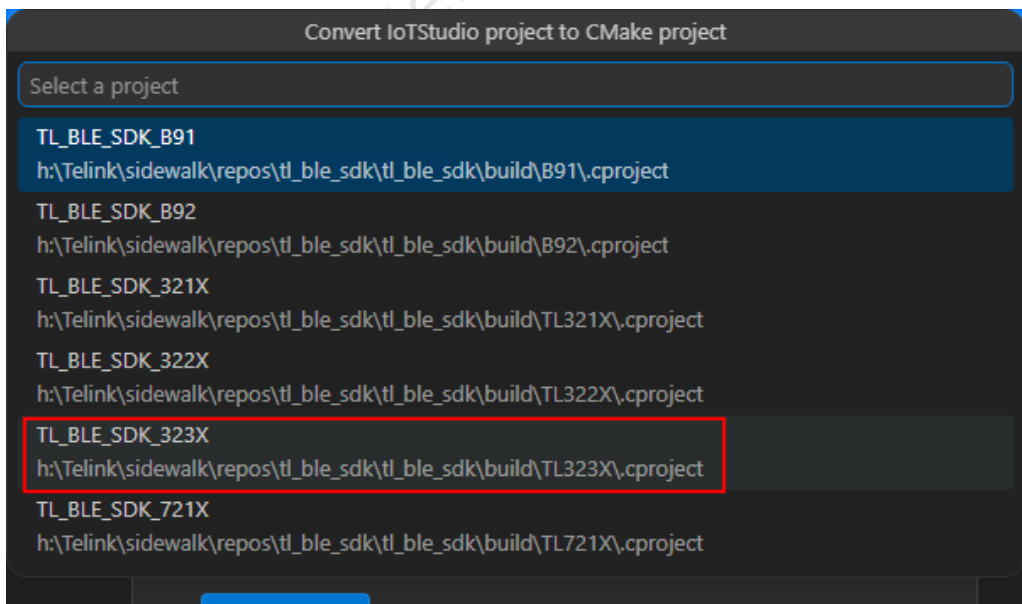


Figure 4.2: TL3238E1_selection

Build the desired build variant (for example, Amazon_sid_dut) by clicking the rocket button next to the build variant name.

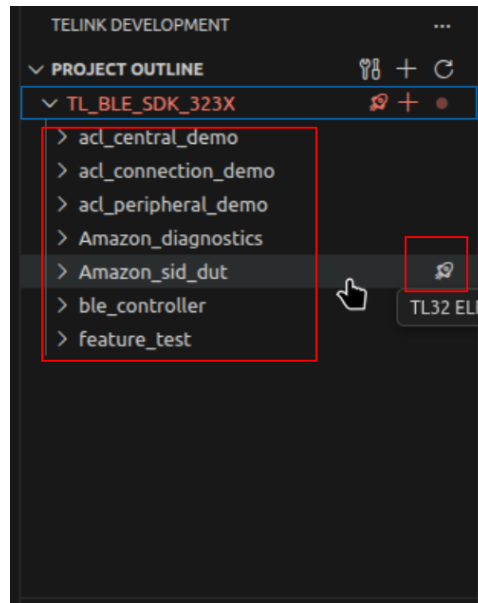


Figure 4.3: `start_build`

The build process should start, and the build artifacts should be visible on final.

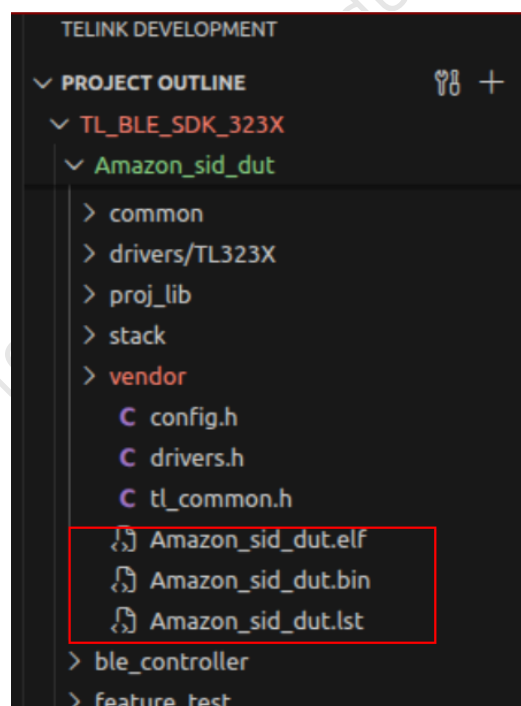


Figure 4.4: `build_artifacts`

4.4 Prepare MFG file

The MFG blob is required to authenticate the Sidewalk end device with the cloud infrastructure.

The MFG blob is generated during the device provisioning process.

Note:

- The Amazon reference app provides a device provisioning script, but it uses the old offset-based MFG blob format. To use the new TLV-based format, some additional steps are required, as shown below.

Locate the provision folder within the Telink SDK:

```
./tl_ble_sdk/vendor/Amazon_sidewalk/sidewalk/apps/common/sidewalk_sdk/tools/provision
```

Convert it to an absolute path (adjust according to your SDK location):

```
/home/user/tl_ble_sdk/vendor/Amazon_sidewalk/sidewalk/apps/common/sidewalk_sdk/tools/provision
```

Go to `aws-iot-core-for-amazon-sidewalk-sample-app` and edit `config.yaml`.

Put the path from the previous paragraph into `PROVISION_SCRIPT_DIR`:

Config:

```
AWS_PROFILE: default # Name of your AWS profile from .aws/credentials
DESTINATION_NAME: SensorAppDestination # Sidewalk destination used for uplink traffic
routing
HARDWARE_PLATFORM: NORDIC # Available values: NORDIC, TI, SILABS or ALL
USERNAME: user
PASSWORD: *****
INTERACTIVE_MODE: False
```

Outputs:

```
DEVICE_PROFILE_ID: *****-****-****-****-*****
WEB_APP_URL: *****.cloudfront.net
```

_Paths:

```
PROVISION_SCRIPT_DIR:/home/user/tl_ble_sdk/vendor/Amazon_sidewalk/sidewalk/apps/common/
sidewalk_sdk/tools/provision
SILABS_COMMANDER_TOOLS_DIR: null # Not needed if Silabs Commander is already in system
Path. Only needed for SILABS.
```

Start the device provisioning script:

```
aws-iot-core-for-amazon-sidewalk-sample-app$ python EdgeDeviceProvisioning/
provision_sidewalk_end_device.py
```

After the script execution is complete, navigate to:

```
./aws-iot-core-for-amazon-sidewalk-sample-app/EdgeDeviceProvisioning/DeviceProfile_[device
Profile]/[device_ID]/
```

The MFG.bin file should be located there. Keep this file for further steps.

4.5 Flash the Board

4.5.1 Address Map

Memory layout for the TL3238E1 (ML3238S module) with 1024 KB flash.

Address	Size	Contents	Notes
0x00000000	~412 KB max	Firmware binary	Varies by demo (see below)
0x000F9000	~4 KB	MFG provisioning blob	Device certificates and keys

Note:

- **Breaking change in V1.0.0.6:** The MFG blob address moved from 0x000F9000 to 0x000F5000. If upgrading from an earlier SDK version, update your flashing scripts accordingly.

4.5.2 Flashing Procedure

Connect the AIOT-DK1 board to the PC via the J12 connector, as shown below.



Figure 4.5: AIOT-DK1 and PC Connection

Start the BDT tool.

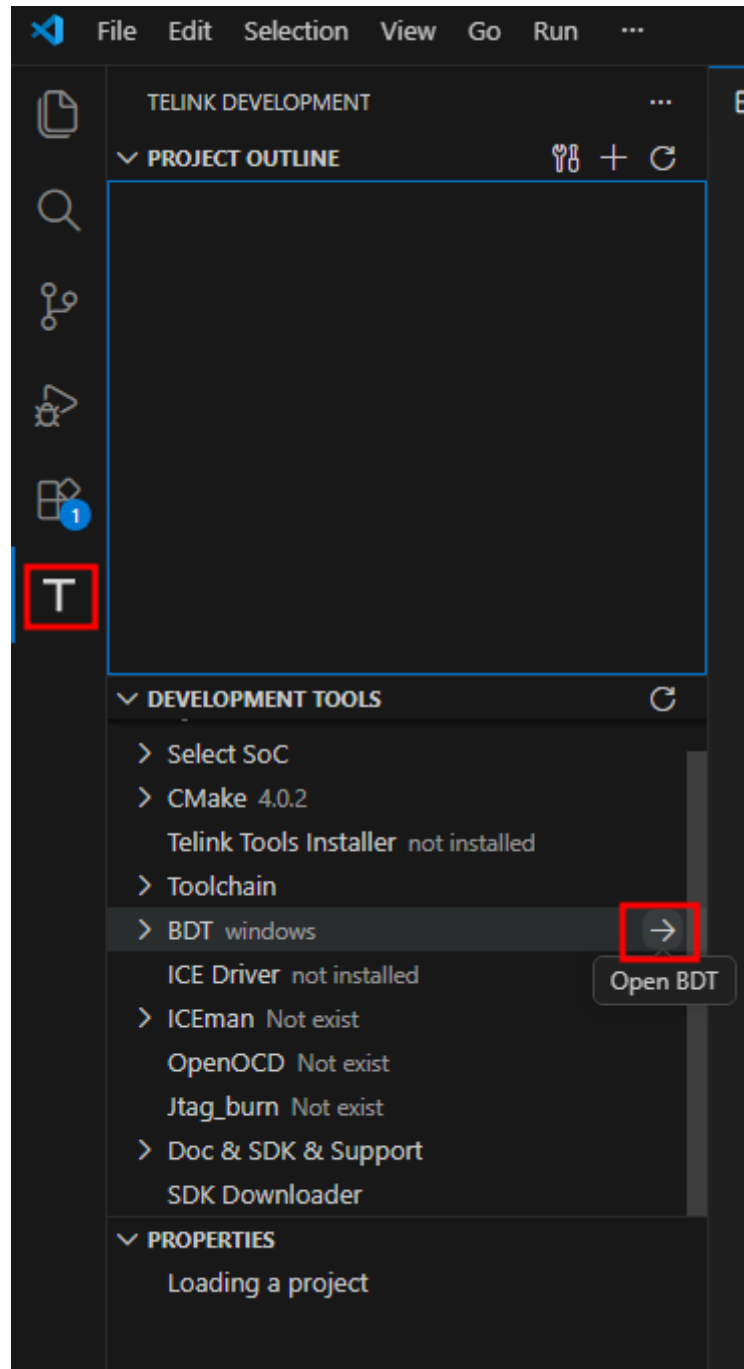


Figure 4.6: Start the BDT

Select the onboard programmer:



Figure 4.7: Select the Onboard Programmer

Make sure the hardware has been detected by the BDT tool. If the panel (1) is empty, click "Install Drivers" (2) and follow the instructions.

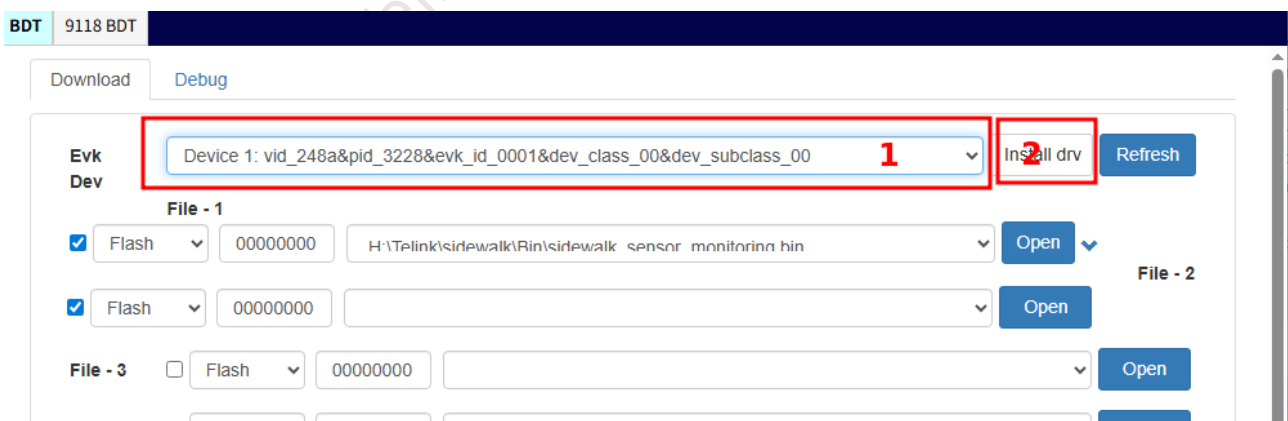


Figure 4.8: BDT Hardware Selection

Ensure the **TL323X** chip family is selected (1), click "Activate"(2), and make sure communication with the target is established (3).

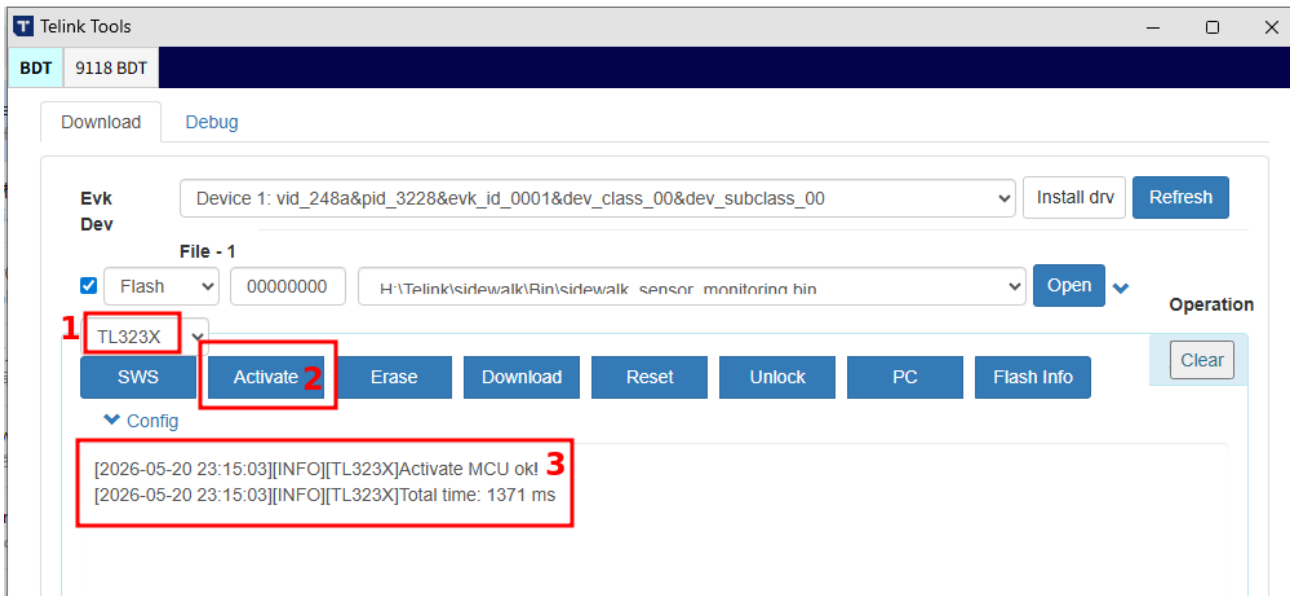


Figure 4.9: BDT Activate

To bring the hardware to a defined and predictable state, it is recommended to erase the entire flash memory. Open the configuration panel (1), enter the flash memory size (2), enable "Unlock flash prior to erase" (3), click "ERASE" (4), and monitor the progress (5).

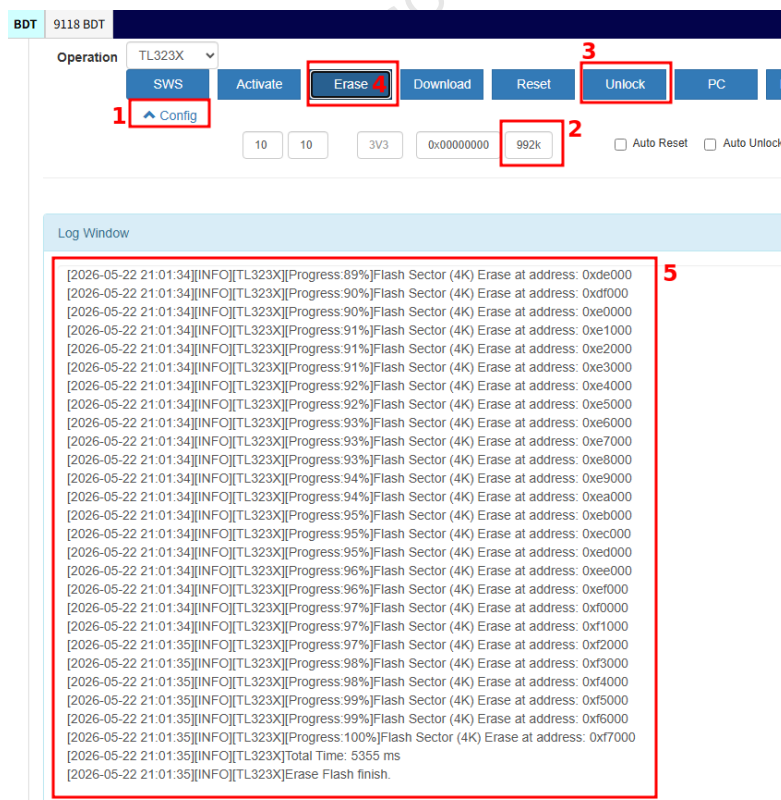


Figure 4.10: BDT Erase

Note:

- Do not erase memory above 0xF8000 (992 KB). Some entries, such as factory calibration data, may be corrupted.

To run the demo application, the flash memory should contain the following two entities:

- Application binary (the actual application, executable machine code)
- MFG binary (contains binary blob)

For the out-of-the-box experience, an application based on Amazon Sensor Monitoring has been prepared. It can be downloaded at [sidewalk_sensor_monitoring.bin](#).

Load the demo application (1) and the MFG blob (2) into the BDT tool. Make sure the download addresses are correct: Application = 0x0000000, MFG = 0x0F9000 (3).

Click "Download" (4) and monitor the progress (5).

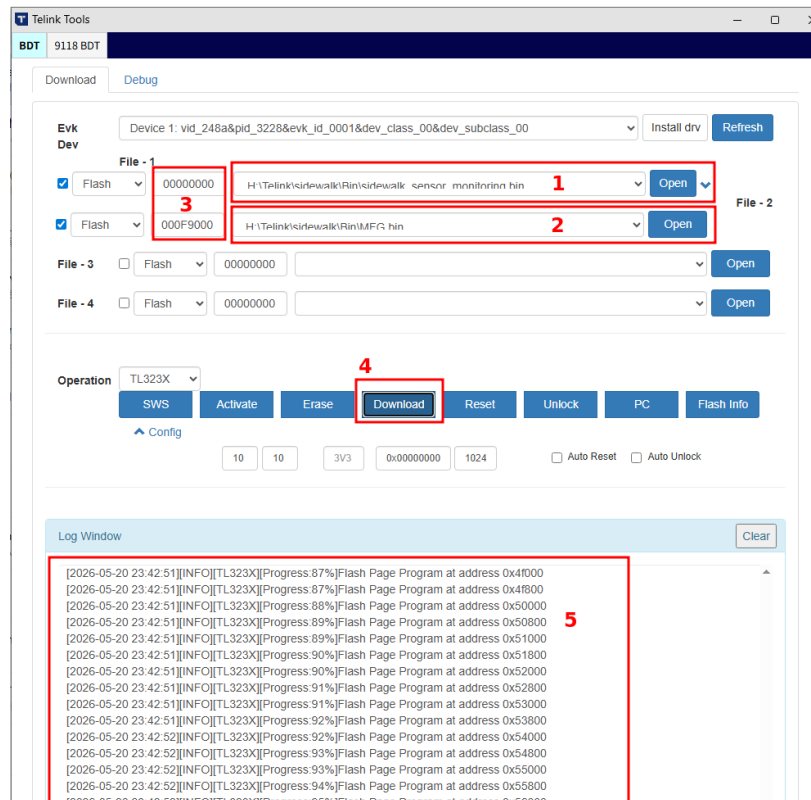


Figure 4.11: Download

5 Sidewalk Demo Test

5.1 Test the Sidewalk Demo

Use the SW2 button (reset) to start the firmware execution.

The D1 and D2 LEDs should light up and go dark when the Sidewalk connection is established. After a while, the web application (using the link obtained in the [Set Up the AWS Cloud](#) section) will detect the device. Temperature reports should become visible, the SW4 button should control the “Push Button Device”, and LED1/LED2 should control the white/green LEDs on the board.

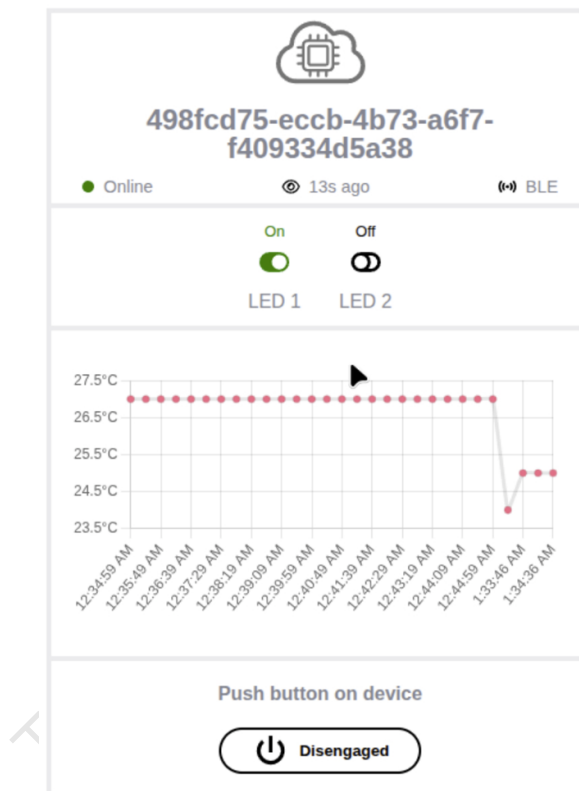


Figure 5.1: Web App

5.2 Test the DUT Demo

The DUT (Device Under Test) demo provides an interactive UART CLI for Sidewalk operations. This is the primary demo for development and testing.

5.2.1 Use Previously Built Demo

Amazon_sid_dut.bin

Refer to [Build the Firmware](#).

5.2.2 Flash

Amazon_sid_dut.bin at 0x00000000

MFG.bin (used in previous demo) at 0x000f9000

5.2.3 Connect UART

Connect the UART-to-USB converter leads (TX/RX/GND) to the J4 pins (mikroBUS header) on the AIOT-DK1 as follows:

GPIO	Function	AIOT-DK1 location	USB Dongle Pin
PA1	TX (board talks)	J4 AN	RX (dongle listens)
PA2	RX (board listens)	J4 TX	TX (dongle talks)
GND	Signal ground	J4 GND	GND (dongle ground)

Note:

- Male pins are required to connect to the mikroBUS header.
- The IO voltage of the TX/RX-related pins here is 3.3V.

5.2.4 Open Terminal

Use the terminal software of your choice (like minicom or realterm, etc.)

Configure on /dev/ttyUSBx or COMx: related to the UART-USB dongle connected.

Use 115200 bps baudrate.

5.2.5 Reset the Board

You should see on the terminal:

```
[000b]sid QA cli application started...
[000c]starting asd_rtos_cli loop
-->
```

5.2.6 Connect to Sidewalk Network

Write a command into the terminal in order to initiate sidewalk stack in BLE mode:

```
sid init 1
```

After a successful response:

```
[001d]CMD: ERR: 0
```

Write a command to register the Sidewalk stack:

```
sid start
```

Wait until successful time synchronization, it should look like:

```
[0077]Updated time: 1457484450.656250
```

5.2.7 Open the AWS IAM Console

Open your AWS IAM console (Amazon account) and navigate to AWS IoT -> MQTT test client. Use “#” wildcard to subscribe to all MQTT topics.

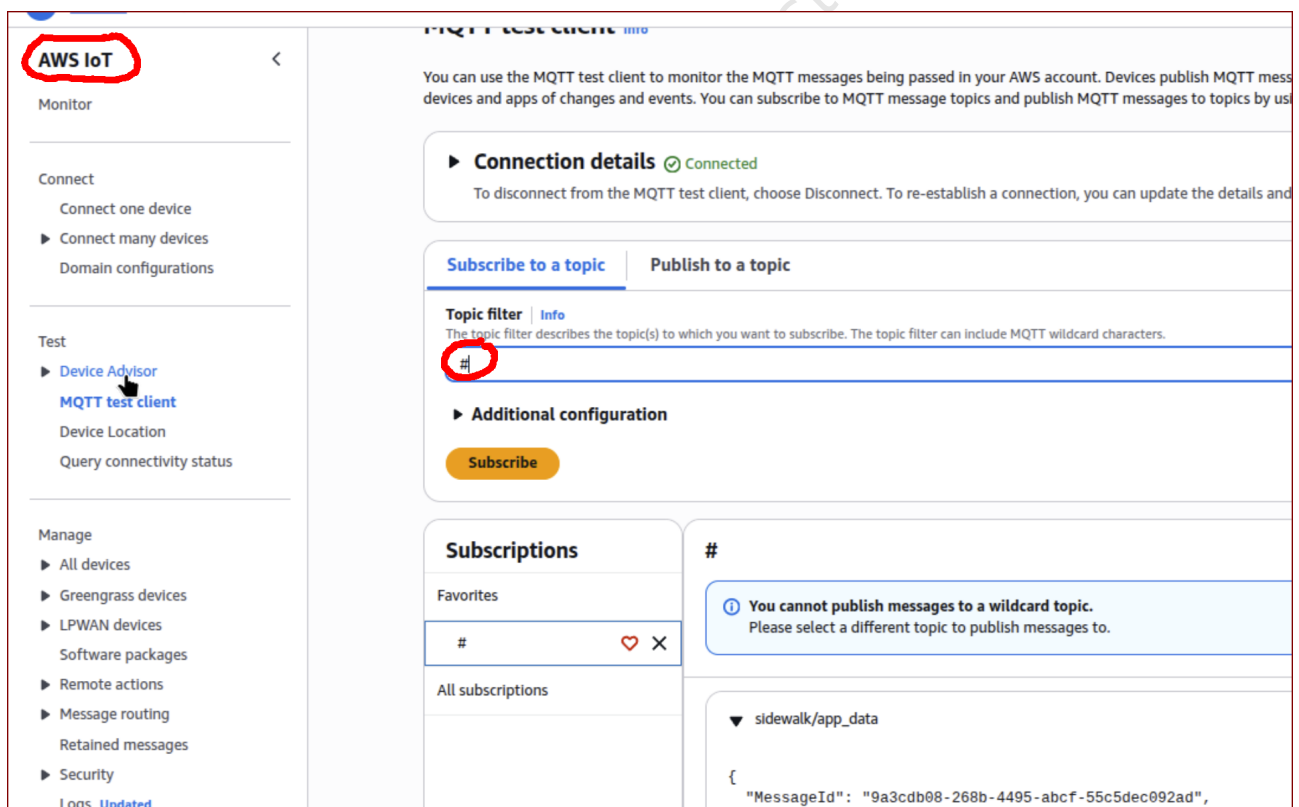


Figure 5.2: MQTT Test Client

5.2.8 Send Test Message

Using the UART console connected to the Telink demo, write a command:

```
sid conn_req 1
```

This should establish a BLE connection between the Telink demo and the gateway.

Wait until the log indicates that the connection is established:

```
[0158]EVENT SID STATUS LINK MODE: LORA: 0, FSK: 0, BLE: 1
```

Issue another command:

```
sid send -a 1 1 30 Hello
```

Go to the MQTT test client on the AWS IAM console:

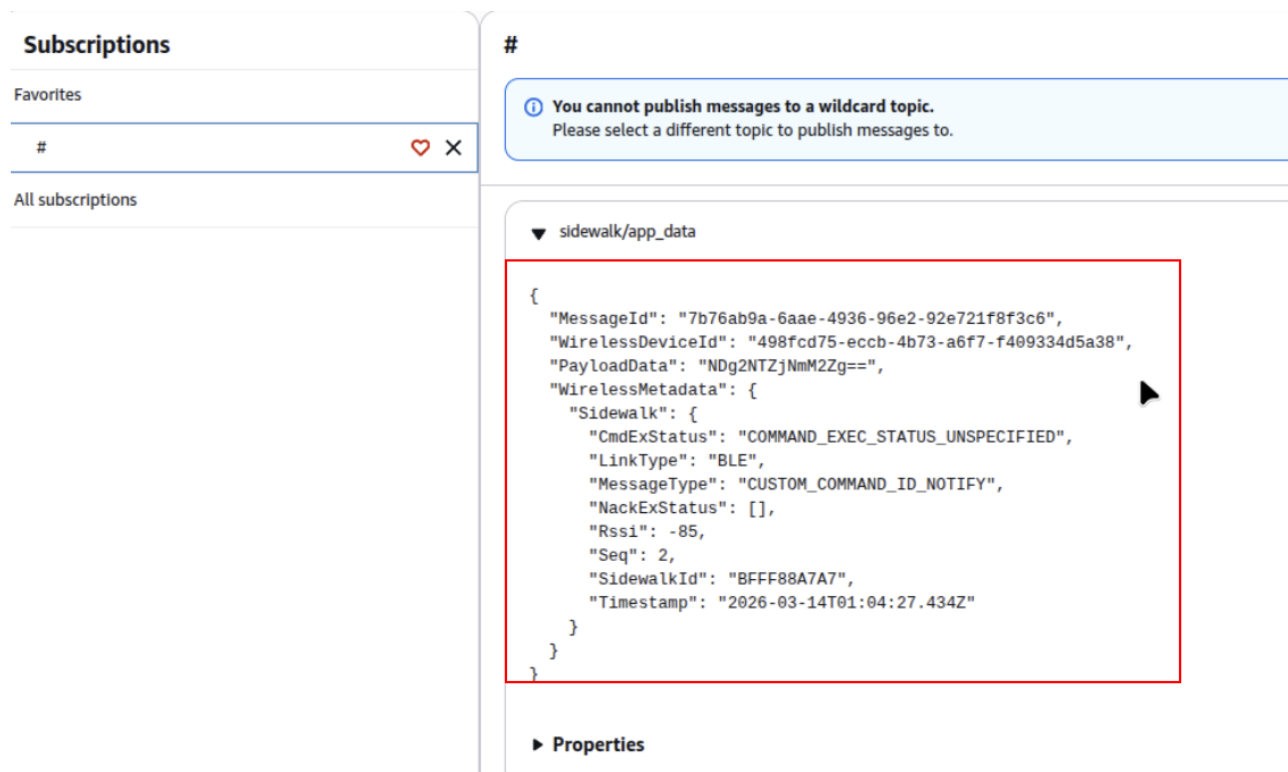


Figure 5.3: MQTT Test Client

```
NDg2NTZjNmM2Zg==
```

You can use the widely available Base64-to-hex conversion to get:

```
34383635366336633666
```

after converting this hex to a string:

```
48656c6c6f
```

And converting to string again:

```
Hello
```

5.2.9 Receive Test Message

Connect the board to the Sidewalk network, like in [Send Test Message](#) from AWS CLI tool (used in web app deployment and provisioning steps), issue a command:

```
aws --region us-east-1 iotwireless send-data-to-wireless-device --id device_id_from_provisioning  
--transmit-mode 0 --payload-data "SGVsbG8gVGVSaW5rIQ==" --wireless-metadata "Sidewalk={Seq=42}"
```

Observe the UART console of the Telink demo board. There should be a log like:

```
[00f0]Data: 48656c6c6f2054656c696e6b21
```

which can be converted to a string.

Note:

- When instead of message data, you find log line looking like this:

```
[02ac]msg dup: 4101000000 (42)
```

It means that message number 42 was already sent. All you need is to increase the number of "Sidewalk={Seq=xx}".

6 Troubleshooting

Common issues and solutions when working with the Telink Sidewalk SDK.

6.1 Flashing Issues

6.1.1 BDT won't Start Flashing or Erasing

Symptom: Clicking Download or Erase in BDT does nothing.

Solution: Use the BDT **Activate** function to wake the MCU, then retry.

6.1.2 Board not Detected During Flashing

Checklist:

- The board must **not** be connected to USB during flashing — power comes through J17 only
- Verify J17 wiring: Red = 3.3V, White = SWS/SWM, Black = GND
- Confirm the correct SoC (TL3238E1) is selected in BDT
- Set flash memory size to **1024K** for ML3238S

6.1.3 MFG Blob not Taking Effect After Reflash

Always **erase the full flash** before writing a new MFG blob. Residual data from a previous MFG can cause provisioning failures.

6.1.4 Wrong MFG Address

The MFG blob address changed in SDK V1.0.0.6:

- **Old address:** 0x000F5000
- **New address:** 0x000F9000

If upgrading from an earlier SDK version, update your flashing configuration.

6.2 VPN Setup for Non-US Development

Amazon Sidewalk is currently available only in the US, Canada, and Mexico. Developers in other regions must route their gateway's internet connection through a US-based VPN.

6.2.1 Proven Setup

- **Router:** GL.Inet GL-MT300N-V2 (Mango)
- **VPN provider:** NordVPN
- **Protocol:** OpenVPN

6.2.2 Steps

- Set up the GL.Inet router with NordVPN using the OpenVPN protocol
- Connect your Echo 4 gateway to the GL.Inet router's Wi-Fi
- The gateway will appear as US-located to Amazon's servers

Note:

- No other VPN provider or protocol has been officially tested. Amazon recommends the OpenVPN protocol.

6.2.3 Sub-GHz in Europe

Sidewalk uses 902–928 MHz for sub-giga (FSK, LoRa). In the EU, this frequency range conflicts with LTE bands protected by regulatory agencies.

For sub-GHz development outside the US:

- Use an RF shielded cage or room
- Alternatively, convert the Echo 4 for cable operation using a u.FL SMA pigtail with a 40–60 dB attenuator (requires RF/hardware skills, not officially recommended)

6.3 AWS Cloud Issues

6.3.1 Deployment Fails with Permission Errors

The `generate_policy.py` script does not generate all required permissions. You need to add these additional inline policies to your IAM user:

- **S3 Bucket Tagging**
 - `s3:PutBucketTagging` , `s3:GetBucketTagging` , `s3:DeleteBucketTagging`
- **IoT Wireless Tagging**
 - `iotwireless:TagResource` , `iotwireless:UntagResource` , `iotwireless:ListTagsForResource`
- **IoT Wireless Full Access**
 - `iotwireless:*`
- **AmazonDynamoDBFullAccess**
 - attach the AWS managed policy

See [Set Up IAM Permissions](#) for the full JSON policies.

6.3.2 MFG Blob from Provisioning Script is Incompatible

The MFG blob generated by `provision_sidewalk_end_device.py` is not directly compatible with the Telink demo due to format differences.

Solution: Use the Telink-specific provisioning script instead:

```
cd tl_ble_sdk/vendor/Amazon_sidewalk/sidewalk/apps/common/sidewalk_sdk/tools/provision/  
python3 provision.py nordic aws --output_bin mfg.bin --certificate_json certificate.json --addr  
0x1F4000
```

6.4 Sidewalk Connection Issues

6.4.1 Device doesn't Connect to the Gateway

Checklist:

- MFG blob is correctly flashed at `0x000F9000`
- Gateway (Echo 4) is online and registered as a Sidewalk gateway
- Gateway and device are within BLE range (~30 ft / 10 m for reliable connection)
- If outside the US: VPN is active on the gateway's network
- Full flash erase was performed before writing the MFG blob

6.4.2 Time Sync Never Completes

After the `sid start`, the device must synchronize time with the Sidewalk network. If `Updated time:` never appears: - Verify the gateway is online and connected to the internet - Move the device closer to the gateway - Check that the MFG blob matches the device profile in AWS IoT

6.4.3 `msg dup` when Receiving Downlink Messages

```
[02ac]msg dup: 4101000000 (42)
```

This means the sequence number has already been used. Each `Seq` value in the `send-data-to-wireless-device` command must be unique. Increment the number:

```
--wireless-metadata "Sidewalk={Seq=43}"
```

6.5 Build Issues

6.5.1 Projects not Visible in VS Code Extension

After opening the SDK in VS Code, if the Telink extension's Project Outline is empty: - Click the **Convert** button in the Project Outline sidebar - Wait a few seconds for projects to populate - Select **TL3238E1** as the target SoC

6.5.2 CMake or Toolchain not Found

When clicking the build (rocket) icon: - If prompted "No find toolchain": click **Configure** and point to your Telink IoT Studio toolchain path (e.g., `.../toolchains/nds32le-elf-mculib-v5f/bin/riscv32-elf-gcc`) - If prompted "CMake is not available": click **Install** and wait for completion (~1 min)

6.6 UART / Debug Log Issues

6.6.1 No Output on UART Console

- **Sensor monitoring demo**: uses PA4 at **1,000,000** baud (1 Mbps) — most terminal apps default to 115200
- **DUT demo**: uses PA1 (TX) / PA2 (RX) at **115,200** baud
- Verify the correct GPIO pins and baud rate for your demo
- Check that TX/RX are not swapped between the board and USB dongle

6.6.2 Garbled Output

Wrong baud rate.

Double-check:

- Sensor monitoring: 1,000,000 bps
- DUT: 115,200 bps

6.7 Known Issues (V1.0.0.6)

- ACL peripheral connection may fail with low probability
- OTA has a small probability of failure
- SBDT demo: abnormal operations during upgrade may cause system exceptions
- Firmware >256 KB requires calling `blc_ota_setFirmwareSizeAndBootAddress()` before `sys_init()`

7 Appendix

7.1 Setting Precise Permissions for IAM Accounts

7.1.1 Set Up IAM Permissions

The IAM account requires the necessary permissions to deploy and run the application. AWS provides a script that generates all required permissions as a JSON policy.

Run the Python script and store its output:

```
python ApplicationServerDeployment/policies/generate_policy.py
```

Open the generated file (DeployStackPolicy.json) and copy its contents to the clipboard.

Go back to the AWS root account console and navigate to IAM -> Policies -> "Create policy" (the yellow button in the top-right corner).

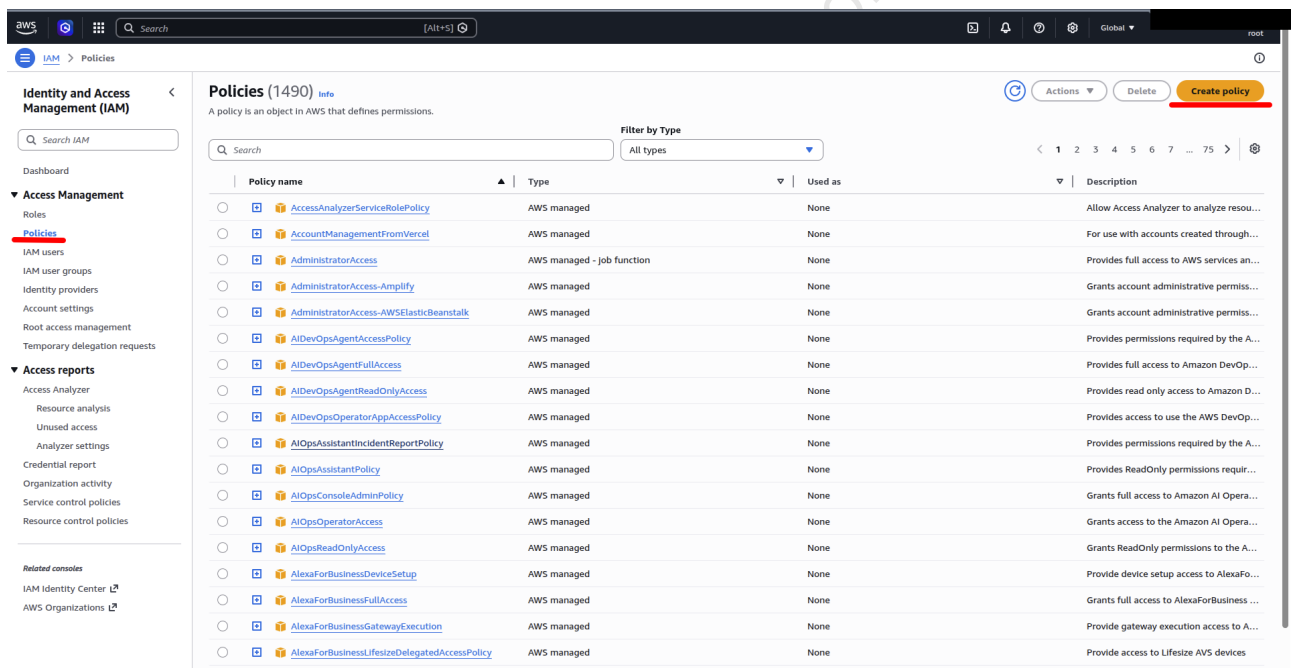
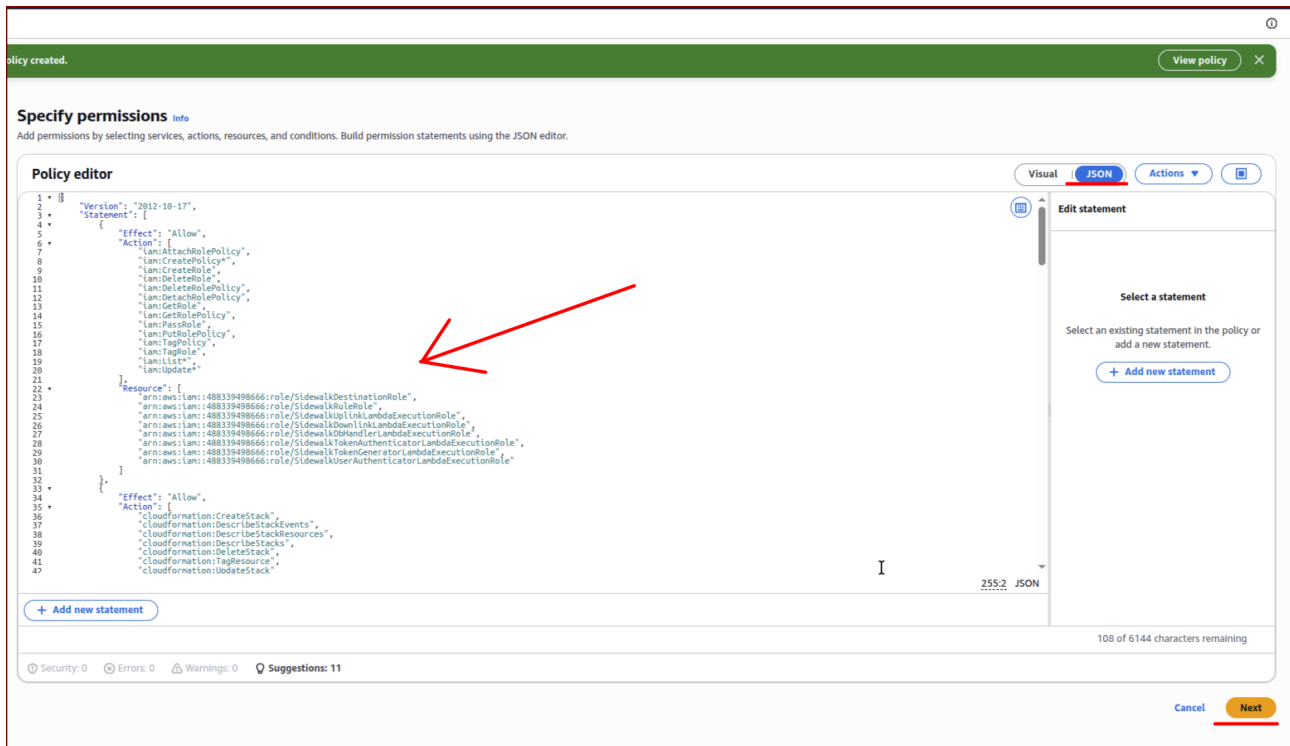


Figure 7.1: Create Policy

Switch to the "JSON" and paste the contents of `DeployStackPolicy.json`.



Policy created. [View policy](#)

Specify permissions info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

Visual **JSON** Actions

1
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3
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42
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44
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46
47

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "iam:AttachRolePolicy",
        "iam:CreatePolicy",
        "iam:CreateRole",
        "iam:DeleteRole",
        "iam:DetachRolePolicy",
        "iam:GetRole",
        "iam:GetRolePolicy",
        "iam:PassRole",
        "iam:PutRolePolicy",
        "iam:TagRole",
        "iam:UpdateRole"
      ],
      "Resource": [
        "arn:aws:iam:488339498666:role/SidewalkDestinationRole",
        "arn:aws:iam:488339498666:role/SidewalkRuleRole",
        "arn:aws:iam:488339498666:role/SidewalkUpLinkLambdaExecutionRole",
        "arn:aws:iam:488339498666:role/SidewalkDownLinkLambdaExecutionRole",
        "arn:aws:iam:488339498666:role/SidewalkHandlerLambdaExecutionRole",
        "arn:aws:iam:488339498666:role/SidewalkTokenAuthenticatorLambdaExecutionRole",
        "arn:aws:iam:488339498666:role/SidewalkTokenGeneratorLambdaExecutionRole",
        "arn:aws:iam:488339498666:role/SidewalkUserAuthenticatorLambdaExecutionRole"
      ]
    }
  ],
  "Effect": "Allow",
  "Action": [
    "cloudformation:CreateStack",
    "cloudformation:DescribeStackEvents",
    "cloudformation:DescribeStackResources",
    "cloudformation:DescribeStacks",
    "cloudformation:DeleteStack",
    "cloudformation:TagResource",
    "cloudformation:UpdateStack"
  ]
}

```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

+ Add new statement

255/2 JSON

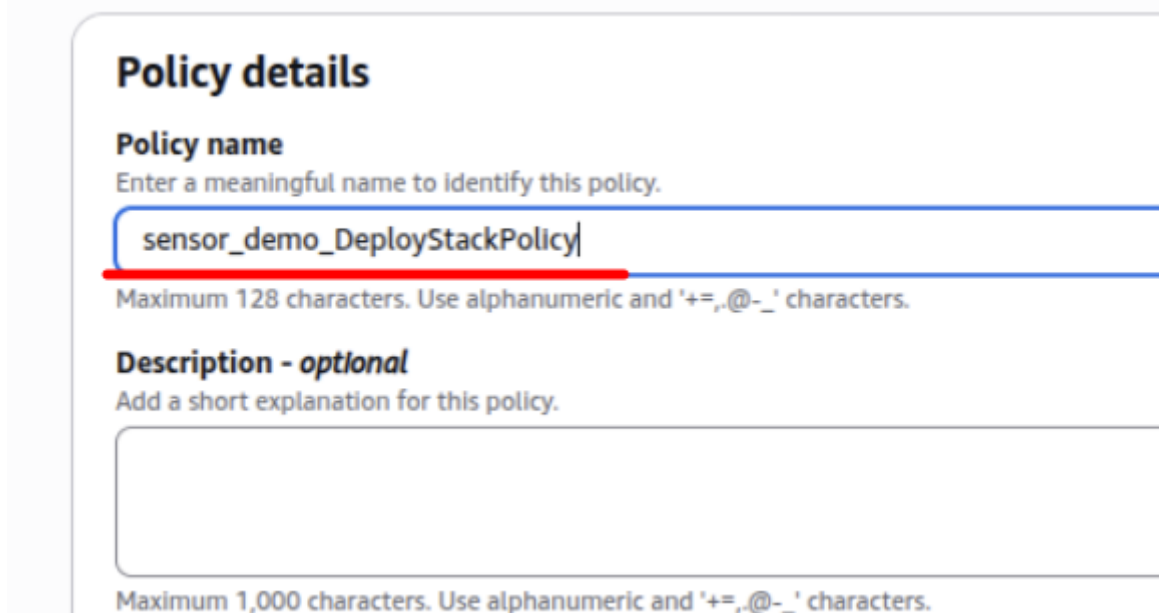
108 of 6144 characters remaining

Security: 0 Errors: 0 Warnings: 0 Suggestions: 11

Cancel Next

Figure 7.2: DeployStackPolicy.json

Enter a meaningful name for the policy, for example, "sensor_demo_DeployStackPolicy".



Policy details

Policy name

Enter a meaningful name to identify this policy.

sensor_demo_DeployStackPolicy

Maximum 128 characters. Use alphanumeric and '+=, @- _' characters.

Description - optional

Add a short explanation for this policy.

Maximum 1,000 characters. Use alphanumeric and '+=, @- _' characters.

Figure 7.3: Enter a Meaningful Name

Click "Create policy" in the bottom-right corner.

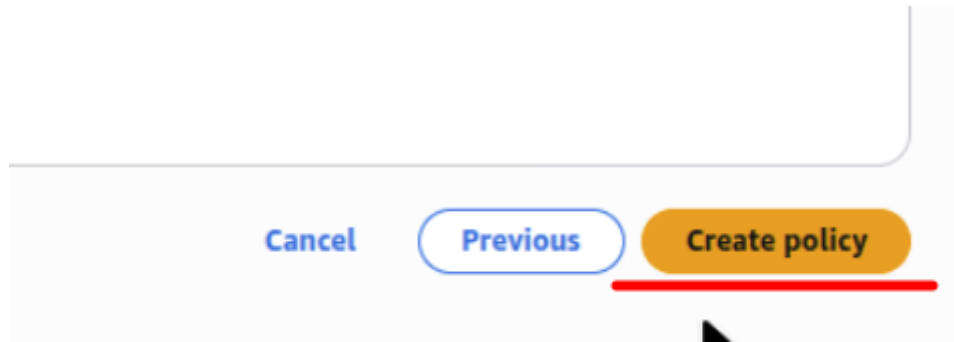


Figure 7.4: Create Policy Button

Add the policy to your IAM user:

While still logged in as the root user, navigate to:

IAM -> Access Management -> Users -> (your IAM user name) -> Add permissions -> Attach policies directly

Search for and attach the newly created policy.

During testing, it turned out that the permissions generated by the script were not sufficient. Additional permissions must also be added.

To add them, navigate to:

IAM -> Access Management -> Users -> (your IAM user name) -> Create inline policy -> JSON

bucket_tagging

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowS3BucketTagging", "Effect": "Allow",
      "Action": [
        "s3:PutBucketTagging", "s3:GetBucketTagging", "s3:DeleteBucketTagging"
      ],
      "Resource": "arn:aws:s3:::*"
    }
  ]
}
```

wireless_tagging

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowIoTWirelessTaggingDestination",
```

```

        "Effect": "Allow",
        "Action": [
            "iotwireless:TagResource",
            "iotwireless:UntagResource",
            "iotwireless:ListTagsForResource"
        ],
        "Resource": "arn:aws:iotwireless:us-east-1:
45[account_ID]54:Destination/ SensorAppDestination"
    }
}

```

wireless_full

```

{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "IoTWirelessFullAccess",
            "Effect": "Allow",
            "Action": "iotwireless:*",
            "Resource": "*"
        }
    ]
}

```

Additionally, grant full access to AmazonDynamoDB:

(From the user panel) -> Add permissions -> Attach policies -> Search for

AmazonDynamoDBFullAccess

7.1.2 Amazon Web Application Deployment

7.1.2.1 Policy Generation Script

```

$ python ApplicationServerDeployment/policies/generate_policy.py
[INFO]      Reading config.yaml file...
[SUCCESS]   File read successfully.
[INFO]      Reading config_grafana.yaml file...
[SUCCESS]   File read successfully.
[INFO]      Fetching AWS account ID based on the caller identity...
[SUCCESS]   Successfully read AWS account ID: 45[account_ID]54.
[INFO]      Personalizing DeployGrafanaPolicyTpl.json template using 45[account_ID]54 AWS
account ID...

```

```
[SUCCESS]      DeployGrafanaPolicy.json policy created.
[INFO]         Personalizing DeployStackPolicyTpl.json template using 45[account_ID]54 AWS
account ID...
[SUCCESS]      policy created.
```

7.2 Using the External Programmer (Black Box)

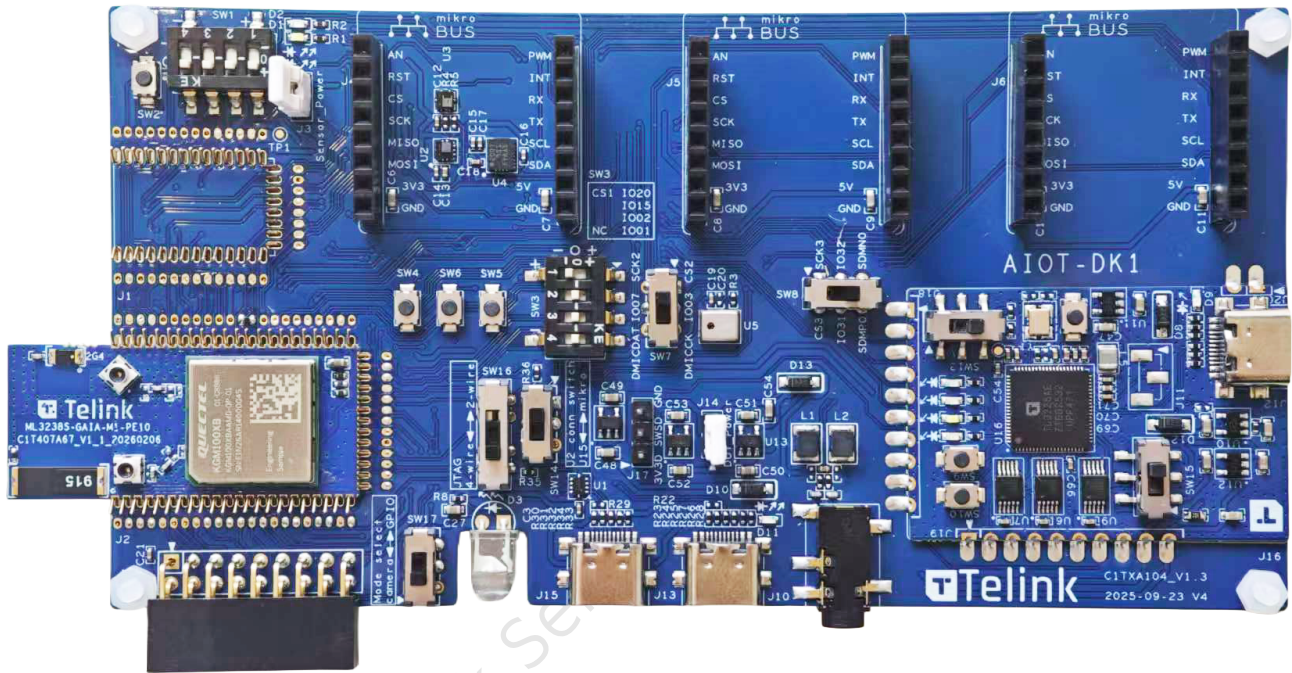


Figure 7.5: AIOT-DK1 Board with ML3238S Module

Attach the ML3238S module to the AIOT-DK1 board. Pay special attention to the pin alignment. We have two main programming methods: one using the Telink programming tool, and the other using the AIOT-DK1 Onboard Programmer (default).

The following demonstration uses the Telink programming tool. For more programming details, please refer to the link: [Burning and Debugging Tool User Guide](#).

Connect the BDT flashing tool to the board using the J17 connector.

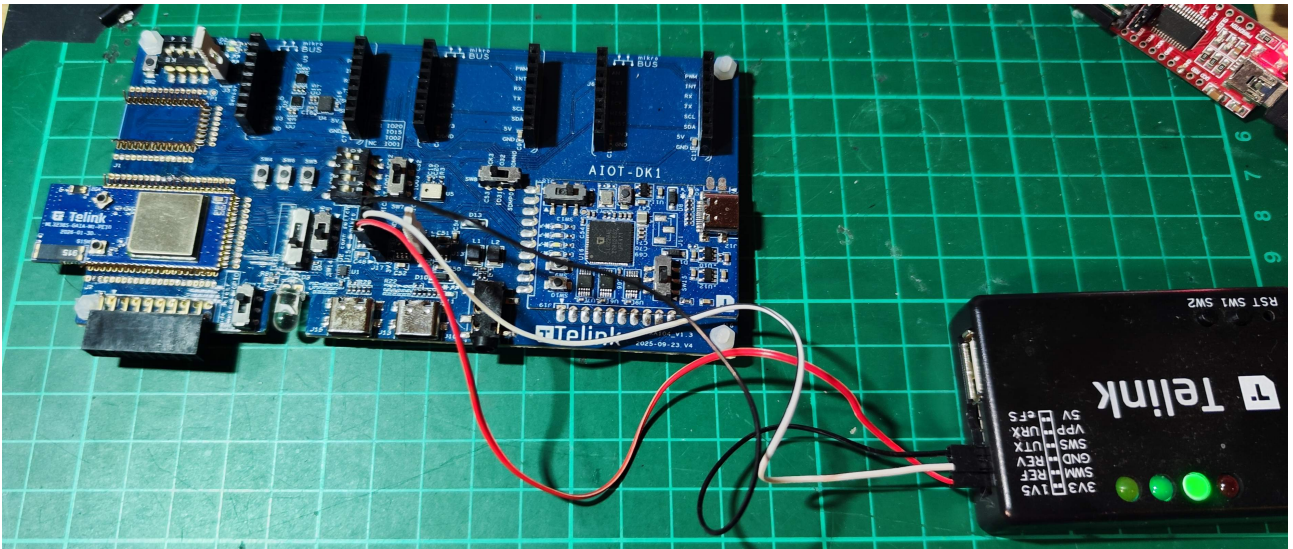


Figure 7.6: Flashing setup

In the picture above:

- RED - 3,3V
- White - SWS/SWM
- Black - ground

Note:

- Use the "Activate" function of BDT too, when flashing/erasing is not starting.

Click on the BDT tool (found in the Development Tools VS Code sidebar) and select the proper BDT device you have.

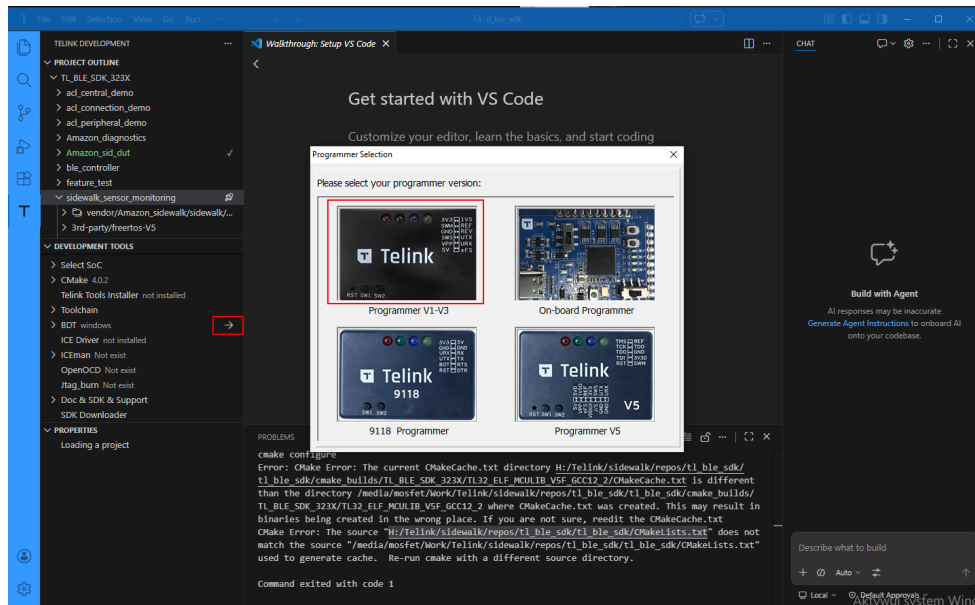


Figure 7.7: Start BDT

Make sure the proper SoC (TL3238E1) is selected:

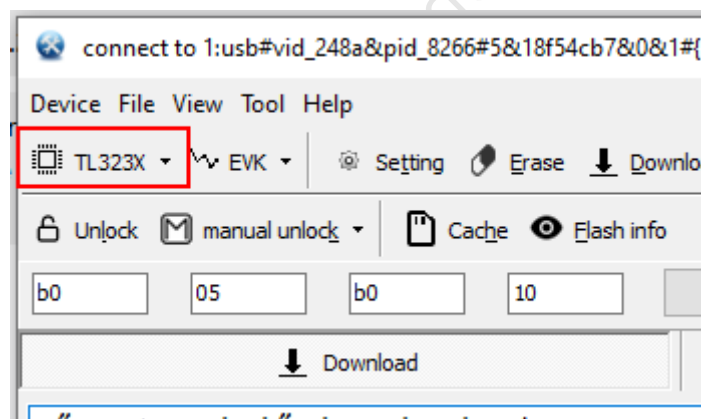


Figure 7.8: SoC Selection

Flash firmware binary alongside the MFG blob.

Firmware binary is flashed to 0x00000000 address.

The MFG blob is flashed to 0x000f9000 address.

To program both the MFG blob ("mfg.bin") and "sidewalk_sensor_monitoring.bin" together, the BDT tool offers multi-address download.

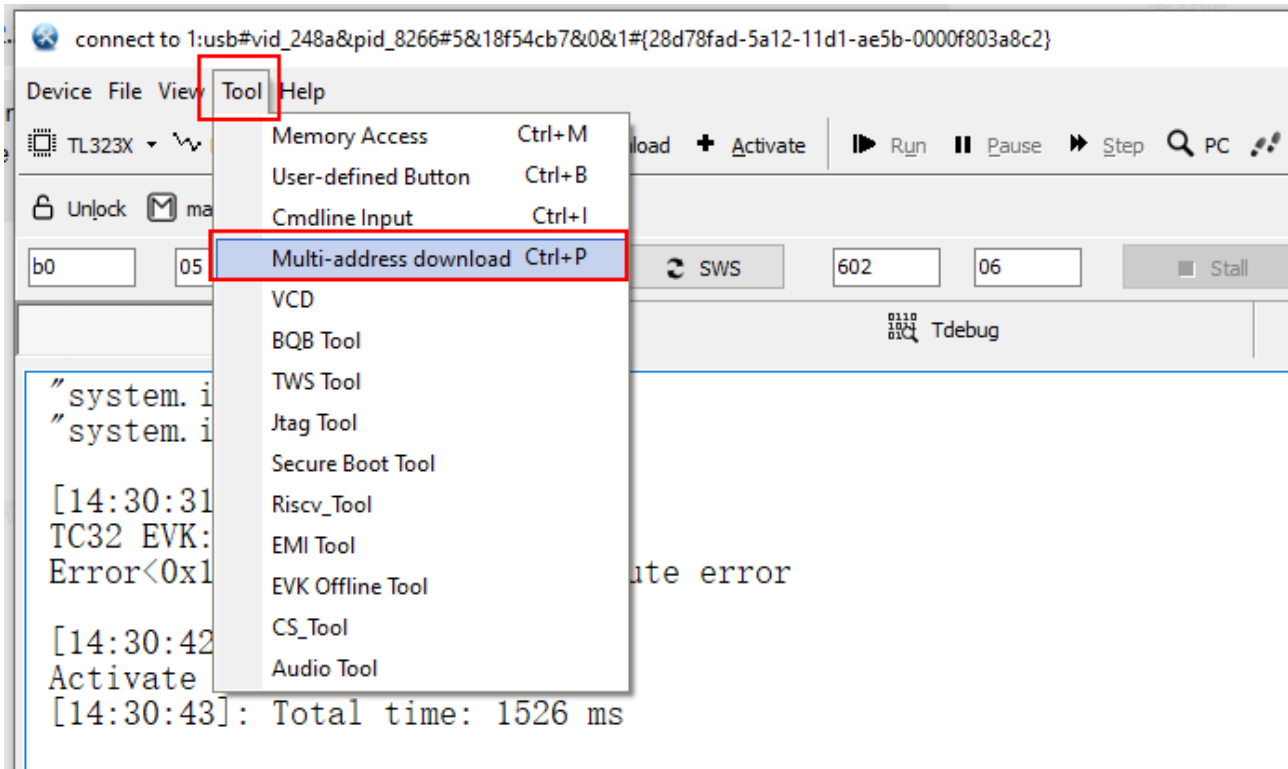


Figure 7.9: Multi Address

Provide paths to the binary and MFG blob and flash the board.

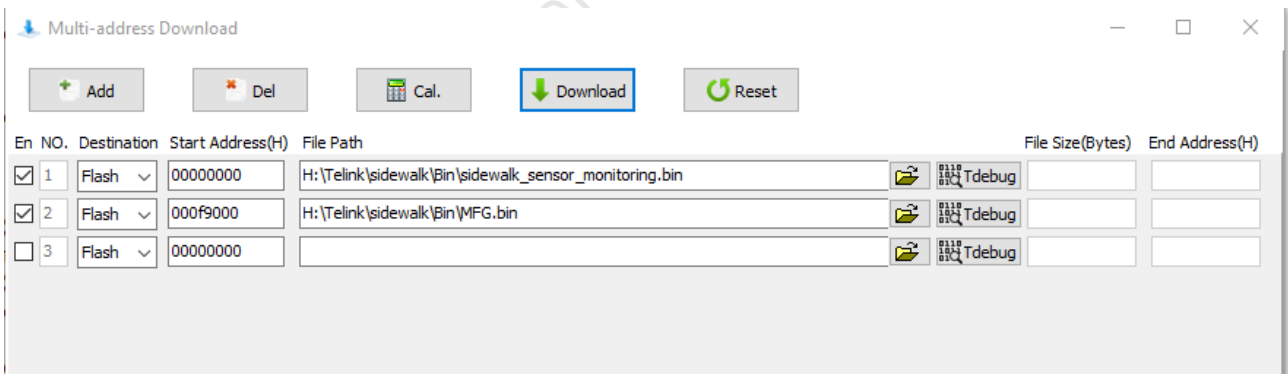


Figure 7.10: Multi Binaries

Hit the "DOWNLOAD" button and wait until flashing is over.

For debugging purposes, the UART->USB converter can be connected to PA4 (available at the "RX" pin J4 "microbus" connector).

Logs are sent at the 1 Mbps baud rate.

Note:

- When uploading a new MFG, it is highly recommended to clean whole flash memory of the Telink SoC.

Make sure proper memory size is configured (1024k in case of ML3238S)

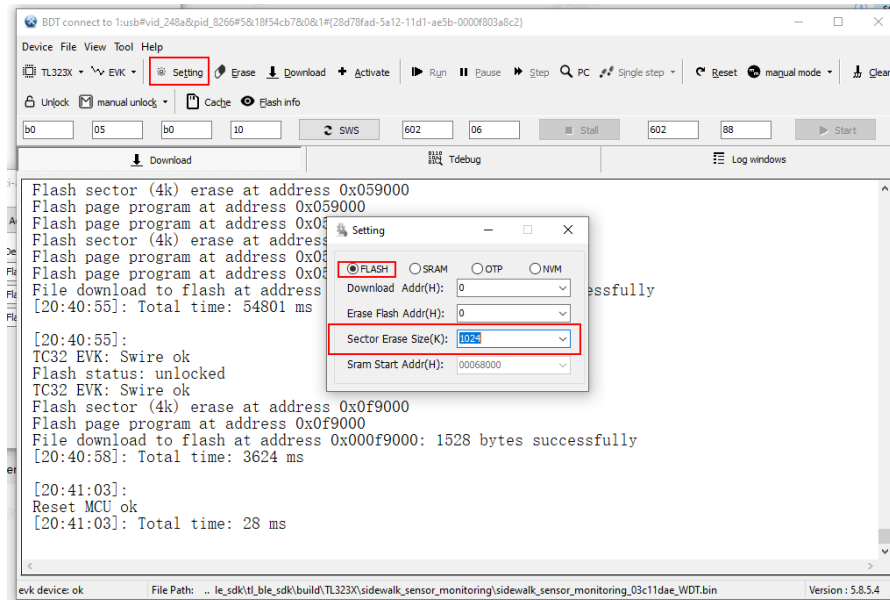


Figure 7.11: Flash Settings

Click "Erase" and wait until the whole memory is cleaned.

Now you can flash using the multi-address download tool:

File	Address
Firmware binary (e.g., <code>Amazon_sid_dut.bin</code>)	<code>0x00000000</code>
MFG blob (<code>mfg.bin</code>)	<code>0x000F9000</code>

Click **DOWNLOAD** and wait for completion.